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Cellular-resolution three-photon ultra-deep imaging (TrUDI) of neuronal population activity at 2-mm depth in the mammalian brain

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Extending the depth reach of multiphoton microscopy for imaging neuronal population activity in the scattering mammalian brain is an 'exponentially hard' problem. Three-photon (3p) microscopy has enabled imaging of neuroactivity at depths up to ~1.2 mm in the mouse brain; however, the fundamental depth limit of 3p imaging, as set by the depth at which out-of-focus background fluorescence exceeds the in-focus fluorescence signal, is well beyond 2 mm. Reaching this fundamental depth limit in 3p imaging has been hampered by insufficient in-focus fluorescence signal. Here we demonstrate a three-photon ultra-deep imaging (TrUDI) system that approaches this fundamental depth limit via systematic and principled optimization of the in-focus fluorescence signal. TrUDI matches the parameters of the excitation laser, the optical system, and the acquisition to linear and nonlinear limits for safe laser exposure while satisfying the spatiotemporal requirements of single-cell resolution calcium (Ca^{2+}) imaging. TrUDI achieves a ~250-fold increase in detected fluorescence photon yield at the same laser power as in previous approaches thereby enabling in vivo cellular-resolution recordings of neuronal population activity within fields of view of ~230 × 230 μm^2 at 8.6 Hz at tissue depths of up to ~2 mm in mouse cortex and hippocampus.

The functional complexity of the mammalian brain arises from its highly interconnected network across regions and depths, each performing specialized roles such as integrating sensory information[1–3], executing parallel computations[4–7] and coordinating behavioral responses[8–16]. These functions are distributed across brain-wide neuronal populations, which communicate through dense, long-range connections[17–20]. The combination of various two-photon fluorescence microscopy (2pM) techniques[21–25] and genetically encoded Ca^{2+} indicators (GECIs) such as the GCaMP variants[26–30] has enabled large-scale recording of neuronal population dynamics at the cellular level in scattering mammalian brain; however, imaging the activity of neurons in its subcortical regions has remained challenging.

The nonlinearity of excitation in 2pM leads to spatial confinement of the fluorescence signal and suppression of out-of-focus fluorescence and thus allows to resolve cellular and subcellular features within the depth range of the mouse cortex; however, the generated fluorescence

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signal in 2pM degrades exponentially as a function of tissue depth due to scattering and absorption loss of excitation photons, as characterized by the effective attenuation length (EAL) of the tissue[31,32]. This imposes a fundamental limit on the depth reach of 2pM[33,34], as this loss cannot be arbitrarily compensated by an increase in the excitation power. At a critical depth (~600–700 µm in the mouse cortex[35,36]) the resulting energy density of the excitation light at the tissue surface will result in a background fluorescence that exceeds the generated signal at the focus, resulting in unity signal-to-background ratio (SBR). While structural 2p imaging at depths exceeding 850 µm has been shown using a combination of longer excitation wavelengths (1,040–1,280 nm)[31,37–40], red shifted static fluorescence indicators and adaptive optics[40], Ca^{2+} imaging using green fluorescence proteins and excitation wavelengths of 920–960 nm has been limited to ~600 µm depth[35,41,42].

In this context, three-photon fluorescence microscopy (3pM)[32,36,43–46] has emerged as a promising solution. The advantages of 3pM stem from two factors: first, the higher-order nonlinearity of excitation in 3pM leads to a substantially lower out-of-focus fluorescence background at the same tissue depth compared to 2pM, and as a result pushes the limit of unity SBR to much greater tissue depths. Second, scattering at the longer wavelength required for 3p excitation is substantially lower, resulting in a longer EAL[47–49].

These advantages, however, come at the cost of a substantially higher required laser pulse energy, both due to a 2–3 orders-of-magnitude lower probability of exciting of a fluorophore in 3p at typical photon densities of ~10^{-34}–10^{-35} photon m^{-2} s^{-1}, and the higher water absorption at typical 3p excitation wavelengths for green indicators such as GCaMP variants[49]. As a result, the imaging depths obtained thus far[36] (~1.2 mm) have been limited either by low fluorescence generated at the focus or laser-induced tissue heating, rather than the fundamental limit, set by a loss of SBR due to out-of-focus fluorescence. Moreover, efforts to maximize the imaging depth within the biological limits of the sample and the photophysics of 3p excitation have been hampered by ad hoc choices of the laser repetition rate, pulse energy and pulse duration, combined with suboptimal pixel excitation schemes.

Here, we demonstrate three-photon ultra-deep imaging (TrUDI), a system which overcomes these limitations and reaches depths that approach the fundamental SBR depth limit of 3pM in the mouse brain. We have developed a 3p acquisition strategy in which excitation parameters of the laser, scanning and the optical parameters of the microscope, are matched to linear and nonlinear limits for safe laser exposure of the sample while satisfying the spatiotemporal requirements of single-cell resolution Ca^{2+} imaging, the photophysics of 3p excitation, and the scattering properties of mouse brain tissue. At comparable imaging depths, our approach yields a ~250-fold increase in detected fluorescence photons at the same laser powers used in previous 3pM approaches.

These innovations have enabled recordings of neuronal population activity at depths up to ~2 mm in mouse brain tissue. We demonstrate this capability of our imaging system by performing imaging throughout the motor cortex and into the striatum, as well as through the posterior parietal cortex (PPC) into the various regions of the hippocampus (HPC), including the CA1 layer, the CA3 layer and the dentate gyrus (DG) in awake mice. We maintained cellular resolution at frame rates of 5–10 Hz within a field of view (FOV) of ~230 × 230 $µm^2$ at a maximal average power level <150 mW out of the microscope objective for our most demanding condition during imaging ~2 mm tissue depth. Using comprehensive immunohistochemical experiments and detailed experimentally informed computational modeling we demonstrate that both our utilized average power levels and pulse energy density conditions are well below the linear and nonlinear safe limits of laser exposure, while showing how power requirements can in principle be further substantially reduced using selective excitation schemes. Thus, TrUDI opens up new possibilities for capturing cellular-resolution dynamics of defined neuronal populations across a tissue column at previously unavailable depths within the scattering mammalian brain.

Results

Depth limit in three-photon microscopy

In multiphoton microcopy—when applied to the mammalian brain—the number of photons at the focus decreases exponentially with depth due to tissue scattering and absorption. To compensate for this loss of photons, and to maintain signal at the focus, the number of photons entering the sample needs to be increased exponentially with imaging depth which causes out-of-focus background fluorescence at the superficial layers. Thus, the depth at which the SBR[23,33,34] reaches unity sets the ultimate achievable depth limit in multiphoton microscopy.

We show the dependence of SBR on focal depth z_f in mouse brain tissue for 2pM at 920–960 nm, based on an EAL 175 ± 25 µm (refs. 32,35,41,42,50) and for 3pM at 1,300 nm for an EAL of 300 ± 50 µm (refs. 42,44,48), with corresponding background-induced depth limits z_{DL} (920–960 nm) = 0.6 ± 0.1 mm and z_{DL} (1,300 nm) = 2.3 ± 0.5 mm (Fig. 1a and Supplementary Note 1). Here, we assumed that fluorescence is generated only by ballistic photons[23,34,35], that the tissue is fluorescently labeled uniformly and that it has uniform scattering and absorption properties across depth, as characterized by a single EAL value. Accordingly, the indicated ranges for the EALs reflect the observed variations from the above assumptions under realistic conditions. Thus, while the above depth reach of 2pM (0.6 ± 0.1 mm) is generally consistent with reported experimental results, a major gap remains between the currently demonstrated depths of 3pM of ~1.2 mm in Ca^{2+} imaging and the fundamental SBR-limited depth reach of ~2.3 mm. As we demonstrate, this gap can be accounted for by the lack of sufficient fluorescence signal at the focus as a result of suboptimal excitation schemes and limitations set by laser-induced tissue heating.

Optimal excitation source and scanning parameters to reach the SBR-limited depth in of 3pM

Traditionally, 2pM but also a number of instantiations of 3pM have employed high-repetition-rate femtosecond lasers. Combined with the typical scan speed of galvanometric mirrors, this resulted in a condition where each image pixel is excited by multiple laser pulses. In this regime, the generated fluorescence signal scales linearly with the number of pulses per pixel. As we have previously shown[36,41,51,52], using a laser at a lower repetition rate (f_{R2}) to deliver the sum of the energy of p pulses of a laser with a higher repetition rate f_{R1} onto a single pixel leads to an effective increase in generated fluorescence signal by a factor $p^2 = f_{R1}^2/f_{R2}^2$ – in other words, a p times higher signal-to-noise ratio (SNR) at the same average power (Supplementary Note 2). Thus, a single-pulse-per-pixel excitation strategy allows for a more efficient use of the total available safe average power budget at the sample, thereby enabling an increase in FOV, acquisition frame rate, or in the case of 3pM, an increase of in-focus fluorescence signal at depth[36,41]. A few recent studies have utilized higher-pulse-energy, lower-repetition-rate lasers in the 0.4–1-MHz regime[32,35,36,43,44,53,54], including some with the above single-pulse-per-pixel excitation scheme[36]; however, the exact choices of laser repetition rate and acquisition parameters seem to be somewhat ad hoc, rather than optimized systematically for maximizing the obtainable imaging depth under safe biological imaging conditions. In the following, we discuss a principled approach that informed our bottom-up engineering and optimization scheme of all acquisition and laser parameters.

In the absence of noise, the SBR, and thus the obtainable depth limit, is independent of the pulse energy; however, in reality various types of noise (biological, readout and shot noise) require that the SNR for both the baseline fluorescence of neurons and their activity transients exceed a certain minimum, as discussed below. First, to detect the baseline fluorescence reliably, the noise of the measurement system (shot noise and systems noise; any noise associated

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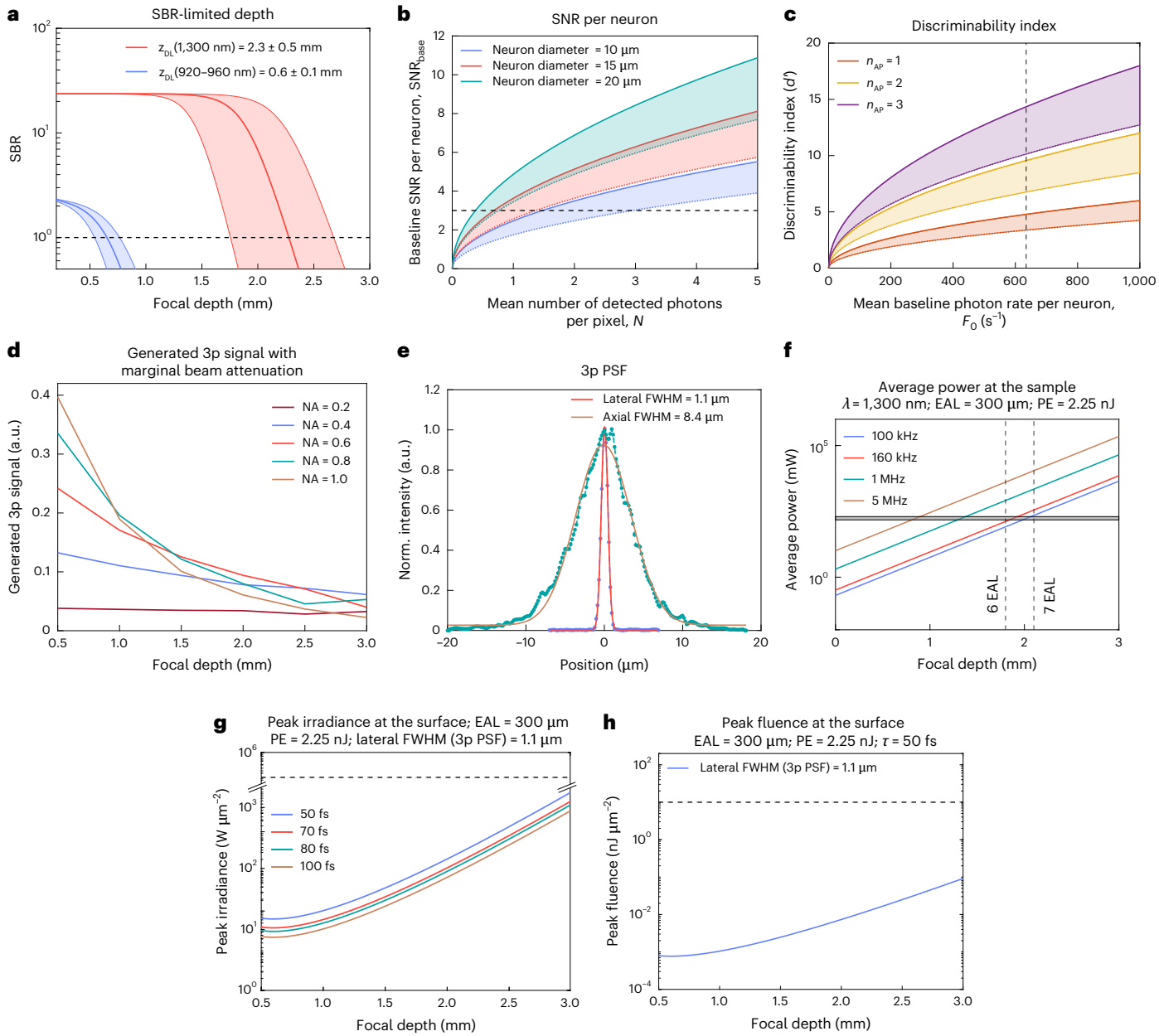


Fig. 1 | Establishing optimal parameters for background-limited deep-tissue three-photon imaging. a, SBR as a function of imaging depth for 2pM at 920–960 nm (EAL 175 ± 25 µm) and 3pM at 1,300 nm (EAL 300 ± 50 µm) in mouse brain tissue, assuming uniform labeling and attenuation properties with expected SBR-limited imaging depth ~0.6 ± 0.1 mm for 2pM and 2.3 ± 0.5 mm for 3pM, where the SBR reaches unity. The shaded region represents the range of EAL variation. **b**, Baseline SNR per neuron (SNR_{base}) sampled by M_{pix} pixels and hence excitation spots per neuron, as functions of the mean number of detected photons per pixel at the PMT, where fluorescence photons in each pixel are generated by a single pulse. The SNR plots are shown for three different neuron sizes (different colors for 10, 15 and 20 µm diameter) for both conditions where background is negligible (SBR » 1, top solid lines of the shaded regions) as well as at SBR-limited depth (SBR = 1, bottom dotted lines of the shaded regions). The horizontal black dashed line represents the minimum baseline SNR criterion of 3 per neuron. **c**, Discriminability index (d'), a measure of neuroactivity SNR, as a function of mean baseline photon rate per neuron (per second) for different numbers of action potentials (n_{AP}), n_{AP} = 1–3. The plots of d' are shown for both conditions where background is negligible (SBR » 1, top solid lines of the shaded regions) as well as at SBR-limited depth (SBR = 1, bottom dotted lines of the shaded regions). d' provides a measure of the accuracy of neuronal activity event detection in terms of TPR and FPR for SBR-limited depth. The vertical dashed line represents the mean baseline photon rate per neuron sampled with 25 pixels at a rate of 8.6 Hz to achieve a minimum baseline SNR threshold of 3 per neuron. **d**, Generated 3p fluorescence signal as a function of focal depth for different excitation NAs. The fraction of ballistic photon reaching the focus and contributing to signal generation was modeled using Monte-Carlo simulation of photon propagation followed by nonlinear 3p fluorescence signal generation, assuming an EAL of 300 µm at 1,300 nm and a paraxial objective lens with 12.5 mm focal length. **e**, Measured lateral (1.1 µm) and axial (8.4 µm) FWHM of 3p PSF for the optimized deep-tissue excitation using 0.5 µm green fluorescence beads at 1,300 nm. **f**, The required average power for 3pM as function of imaging depth for different laser repetition rates for a fixed pulse energy (PE) of 2.25 nJ at the focus, a typical value for generating sufficient fluorescence photons and assuming an EAL of 300 µm. The horizontal gray band denotes the average power range between 150 and 200 mW. The region between vertical dashed lines denotes the focal depths in the range of 6–7 EAL showing the relationship with average power requirement for different laser repetition rates for the fixed 2.25-nJ pulse energy at the focus. **g**, The peak irradiance at the surface as a function of focal depth for a PE of 2.25 nJ, a PSF size as shown in **e**, and an assumed EAL of 300 µm at 1,300 nm. Different colored plots correspond to four different pulse durations. The horizontal dashed line corresponds to the peak irradiance limit of ~2.35×10^5 W µm^{-2} for safe exposure[44]. **h**, The peak fluence at the surface as a function of focal depth for the PSF size shown in **e**, PE of 2.25 nJ, and pulse duration of 50 fs (FWHM), assuming an EAL of 300 µm at 1,300 nm. The horizontal dashed line corresponds to the peak fluence limit of 10 nJ µm^{-2} for safe exposure[44].

with the signal detection process) needs to be overcome. To estimate the minimum number of photons per neuron per second required to achieve a baseline SNR (SNR_{base}) of 3, we considered the typical range of mouse neuron soma diameters (~10–20 µm)[55–57], as well as the need for their faithful spatial sampling and our inferred optimal choice of point spread function (PSF) (see below). Together with the single-pulse-per-pixel excitation principle, these dictate a minimum spatial sampling with ~5 pixels in each dimension, or ~25 pixels in total, per neuron. We assumed Poisson-distributed photons at the detector at a constant mean rate and included an additional noise component that is contributed by the photomultiplier tube (PMT) gain in the regime of low average photon numbers. This allowed us to establish the relationship between the anode current, its variance (or equivalently, pixel digital count and count variance) and the mean number of detected photons (Supplementary Note 3). For typical parameters of our system that include PMT gain, cathode radiant sensitivity and fluorescence response width, we show the SNR_{base} of a neuron as functions of mean number of detected photons per pixel for different neuron sizes, for both in the absence of fluorescence background as well as at the SBR limit of 1 (Fig. 1b). The required SNR_{base} threshold and the corresponding variance of the PMT output current per pixel are shown as functions of mean number of detected photons per pixel for the desired $SNR_{base} = 3$ per neuron (Extended Data Fig. 1a,b). For our smallest (largest) assumed neuron size of ~10 µm (20 µm) diameter, sampled by 25 (100) pixels, the required baseline pixel SNR ($SNR_{base,pix}$) is ~0.6 (0.3) per frame, respectively. At the SBR-limited imaging depth, this corresponds to ~3 (0.8) detected photons per pulse or to ~635 detected photons per neuron per second (irrespective of the neuron size) at our chosen frame rate of 8.6 Hz (Supplementary Note 3), while for the depth range ($\lesssim$1 mm), where the fluorescence background is negligible, this requirement drops to ~320 photons per neuron per second.

The discrimination of neurons' Ca^{2+} activity from their baseline fluorescence can be characterized by the discriminability index (d') given by $d' = \sqrt{\frac{1}{1+1/SBR}}\frac{\Delta F}{F}\sqrt{\frac{F_0\tau_{Ca}}{2}}$ (refs. 42,58), where $\frac{\Delta F}{F}$ is the relative fluorescence change induced by one action potential (AP), F_0 the baseline fluorescence photon detection rate and τ_{Ca} the half-life of the Ca^{2+} decay (Supplementary Note 4). If multiple APs occur within τ_{Ca}, the observed $\frac{\Delta F}{F}$ will scale approximately linearly with the number of APs (n_{AP}) for small (~1–3) n_{AP} (ref. 30). Thus, the AP-dependent discriminability index can be written as $d'(n_{AP}) = \sqrt{\frac{1}{1+1/SBR}}n_{AP}\frac{\Delta F}{F}\sqrt{\frac{F_0\tau_{Ca}}{2}}$. We note that d' is proportional to the z-score (obtained by subtracting the baseline mean from the raw fluorescence time trace and dividing the result by the baseline s.d.), another measure of neuronal activity. Both d' and the z-score depend on the baseline SNR_{base} via the factor $\sqrt{F_0}$. We evaluated d' (n_{AP}) under the conditions that we had previously determined to yield a baseline SNR_{base} greater than 3. At the background-limited depth reach (SBR = 1), using the in vivo parameters of GCaMP8s[30] ($\frac{\Delta F}{F} = 0.6$ and $\tau_{Ca} = 0.2$ s) and at our imaging frame rate, we found d' (n_{AP}) to be 3.4, 6.8 and 10.1 for 1, 2 and 3 APs respectively. This corresponds to a Ca^{2+} transient detection rate (true positive rate) between > 95.5–99.9%, and a false positive rate of <4.5–0.1% for 1–3 APs, respectively. We show the discriminability index for 1–3 APs as a function of the detected mean baseline photon rate per neuron (Fig. 1c), and the corresponding true positive rate (TPR) and false positive rate (FPR) (Extended Data Fig. 1c,d and Supplementary Note 4). Thus, we found that the requirement of $SNR_{base} = 3$ was sufficient to ensure that the requirement of $d' = 3$ for detecting Ca^{2+} transient events at the SBR-limited depth is fulfilled.

Next, we aimed to identify the optimal size of the excitation PSF that maximizes the number of generated photons for a given pulse energy. Traditionally, using the highest possible numerical aperture (NA) for both excitation and photon collection has been assumed to maximize signal under non-saturation conditions; however, when imaging neuronal cell bodies at depth in scattering tissue, this assumption requires reevaluation with respect to the excitation NA, for two reasons: First, the signal generated by the excitation PSF is proportional to the cube of the photon flux and linearly proportional to the axial extent of the PSF in 3pM. Thus, a larger axial PSF, as long as it does not exceed the average diameter of neuronal cell bodies, could enable integration of the fluorescence signal over a greater number of fluorophores. It has been shown that for neuronal imaging, an axial extent comparable to the size of the cell body diameter (~10–15 µm) is preferable[34,36]. Second, a higher excitation NA means that the peripheral rays have a longer beam path to the focal point. This translates to a greater degree of scatter-induced loss of excitation photons at the focus and therefore a reduction of fluorescence signal generation in 3p with increasing imaging depth compared to a lower excitation NA. The generated 3p signal as a function of focal depth without this effect is presented for excitation NAs of 0.2–1.0, showing the expected increase in 3p signal as function excitation NA for constant pulse energy at the focus, independent of imaging depth (Extended Data Fig. 1e). We then took the effect of scatter-induced loss of photons for marginal rays into account by performing Monte-Carlo simulations to estimate the fraction of ballistic photons that reached the focus and contributed to the generated 3p fluorescence signal, as a function of focal depth and for different excitation NAs, assuming the same pulse energy at the tissue surface (Fig. 1d). We found that at lower tissue depth ($\lesssim$1 mm), the traditional expectation that the generated fluorescence signal is maximized for the highest excitation NAs holds; however, at greater tissue depths the optimal excitation NA was lower, as a result of a higher probability of loss of ballistic photons in the marginal rays due to an effectively longer travel path of excitation photons, and hence higher nonlinear falloff of the generated fluorescence signal, at larger NAs. Specifically, we found that (consistent with similar previous findings[39,59,60]) the optimal excitation NA at the SBR-limited depth of ~2–3 mm in the mouse brain is ~0.4–0.6. Notably, while this lower NA range leads to optimal excitation, it is still beneficial to use the full available NA of a high-NA objective for collection (NA 0.8 in our case) as discussed further below.

In addition, we note that the beam diameter at the surface scales as ~NA × 3 for ~2-mm imaging depth (beam diameter ~ $2z_f\sin\theta$, where $\sin\theta = NA/n_{idx}$; $z_f = 2$ mm, the imaging focal depth and $n_{idx} = 1.33$, the refractive index). Thus, signal attenuation is indirectly affected more strongly at higher NAs by factors such as surface inhomogeneities, larger blood vessels and aberrations. Based on these considerations, we used a lateral 3p PSF size of ~1.1 µm full-width-at-half-maximum (FWHM) and an axial 3p PSF size of 8.4 µm FWHM in our TrUDI design corresponding to an effective excitation NA of ~0.3 which was realized by appropriate underfilling of a 0.8 NA long-working-distance objective (Fig. 1e). To further enhance collection efficiency we selected the optical elements for fluorescence collection between the objective back-focal aperture and the PMT active area to allow for faithful capturing and relaying of rays deviating by maximally ~ 8° from collimation (Supplementary Note 5 and Extended Data Fig. 2) resulting in an estimated total fluorescence collection efficiency of ~11–7% for an isotropically emitting source at the center of FOV and 1.5–2-mm depth (Extended Data Fig. 2d).

As the next step, we sought to identify the required pulse energy based on the above optimal PSF size and fluorescence collection efficiency by considering the number of photons needed per neuron per second to achieve a desired d' for reliable Ca^{2+} transient detection. As shown above, a minimum of 635 detected photons per neuron per second is required to satisfy both $SNR_{base} = 3$ as well as a minimum d' of 3 for detecting Ca^{2+} transient events. This corresponds to ~3.0 detected photons per pulse, assuming sampling of the smallest neurons cell bodies (diameters ~10 µm) with 25 pulses at a frame rate of 8.6 Hz. Using our experimentally informed model for fluorescence generation and collection efficiency estimated above as well as the model for

system noise (Supplementary Note 3) we found that to achieve this, a pulse energy of ~2.25 nJ at the focus is sufficient at our PSF size.

We then corroborated our findings regarding the optimal PSF size and required pulse energy by performing in vivo experiments and evaluating the experimentally obtainable pixel SNR. We found good agreement with our theoretical model for SNR (Supplementary Note 3), both for the experimental variance of pixel photon count (Extended Data Fig. 3a) and the pixel SNR (Extended Data Fig. 3b) as a function of mean pixel count. Furthermore, the histogram of our experimentally obtained pixel SNR (Extended Data Fig. 3c) for ~2 nJ pulse energy at the focus demonstrates that our criterion for the baseline SNR per pixel (Extended Data Fig. 1a) was satisfied. These findings are consistent with previous reports of 3pM in which pulse energies of ~1–5 nJ at the focus (for ~40-fs pulses) resulted in sufficient fluorescence signal and SNR for reliable Ca^{2+} imaging while remaining well below the ~10 nJ μm^{-2} peak fluence threshold for nonlinear damage[44].

Next, assuming EAL 300 µm and 50-fs pulse duration, we investigated the relationship between laser repetition rate, average power and obtainable depth for a pulse energy of 2.25 nJ at the focus. We show the key roadblock in reaching the fundamental SBR-limited 3pM imaging depth when MHz range repetition rate lasers are used (Fig. 1f): with these repetition rates, it is impossible to reach the SBR-limited depth because delivering the required pulse energy at the focus would make it necessary to exceed previously used maximal average powers in 3pM by more than an order of magnitude. For example, in a previous study that implemented one-pulse-per pixel excitation, to deliver ~3.5 nJ at the focus ~190 mW of power was required for an imaging depth of ~1.2 mm using a ~1 MHz, 70-fs laser[36], while a subsequent study showed the onset of minor immunohistochemical responses already at ~150 mW for an imaging depth of 1 mm (ref. 42). At the utilized repetition rates in these studies (0.8–1 MHz), to reach the depth of 2 mm while keeping a constant pulse energy of 2.25 nJ at the focus, as was used in our case, >1.4 W of average power would be needed. Clearly, to approach the fundamental SBR-limited imaging depth of 3pM at safe average power while delivering sufficient pulse energy at the focus, lower laser repetition rates are required. A 160 kHz laser can deliver a 2.25-nJ pulse at ~2 mm depth while keeping the average power at the surface below 150 mW (Fig. 1f). As the generated fluorescence signal in 3pM is proportional to the cube of the pulse energy, this corresponds to a ~250-fold increase in 3p excitation efficiency at a given average laser power; however, a few additional considerations are required when using such low repetition rate lasers. First, as discussed above, to resolve neurons and their boundaries spatially, each neuron needs to be sampled by ~25 separated spots[36,41,52]. Thus, the pixel spacing needs to be selected appropriately to ensure faithful spatial sampling of each neuron while maximizing the imaging FOV. For the smallest neurons we considered, with a soma diameter of 10 µm, this translated to pixel spacing of ~1.8 µm. Second, reducing the laser repetition rate, while keeping all other parameters constant, inevitably decreases the pixel acquisition rate and thus the frame rate. This necessitates balancing the tradeoff between spatial sampling and FOV with the additional requirement of sufficient temporal resolution for in vivo Ca^{2+} imaging. Third, as the pulse energy increases, the peak fluence and, depending on the pulse duration, the peak irradiance can in principle exceed the safety limits for nonlinear laser exposure at the tissue surface. Assuming again a pulse energy of ~2.25 nJ at the focus, the resulting peak irradiance and the peak fluence at the sample surface as a function of focal depth are shown (Fig. 1g,h). The safe limits for fluence and irradiance of ~10 nJ μm^{-2} and ~2.3×10^5 W μm^{-2}, respectively[44], would be exceeded at a hypothetical imaging depth of ~5 mm (Extended Data Fig. 4a,b); however, for the ultimate obtainable imaging depth of ~2.3 mm in 3pM, as set by the SBR > 1 requirement (Fig. 1a), both the fluence and the irradiance remain well within safe limits at the surface, and much more so at the focus. Finally, as our identified optimal excitation parameter space allows to remain well within the safe tissue irradiance thresholds for pulse durations over the entire a range of 50–100 fs (Fig. 1g), and as the excitation efficiency in 3pM scales as the inverse squared of the pulse duration[36,42,44] (Supplementary Note 1), we maximized the obtainable fluorescence signal for the same average power by choosing the shortest possible pulse duration of ~50 fs while maximizing the overlap of the resulting spectral full bandwidth of ~85 nm with the 3p absorption band near the absorption peak of GCaMP6f[61].

To implement the inferred laser pulse parameters and optimal excitation conditions for deep 3pM experimentally, we used a customized tunable optical parametric amplifier (OPA) outputting ~50-fs pulse with >10 µJ at 160 kHz. Mapping this repetition rate onto the above requirements for spatial sampling of neurons and the temporal dynamics of GCaMP variants informed the design of our TrUDI system. TrUDI provides access to a FOV of ~230 × 230 μm^2 at a pixel spacing of 1.8 µm while operating under a single-pulse-per-pixel excitation scheme and at a frame rate of 8.6 Hz. Thus, it allows for a substantial increase of obtainable imaging depth in the mouse brain while maintaining cellular and subcellular-scale spatial resolution, within a moderate FOV and at adequate temporal resolution. A schematic of the optical path of the TrUDI is presented (Extended Data Fig. 5).

Experimental comparison of TrUDI with MHz 3pM and in vivo validation of imaging performance

To experimentally validate our optimization ideas, we performed a direct comparison of TrUDI with a state-of-the-art 3pM implementation that used one-pulse-per-pixel acquisition at 1 MHz repetition rate, ~70-fs pulse duration and ~3.5 nJ pulse energy. This reference implementation had previously achieved the deepest 3p Ca^{2+} imaging to date, reaching ~1.2 mm depth at a mean SNR_{base} of ~2.5 (ref. 36). We matched PSF and detection parameters to our TrUDI setup and constructed a similar optical path, with the exception of a longer pulse duration of ~100 fs, necessitated by experimental constraints. We then performed a comparative in vivo imaging in the PPC in mice expressing GCaMP6s (Camk2a-tTA × tetO-GCaMP6s) and sought to reach the same imaging depth in the same animal both using the 1 MHz reference system and TrUDI (Fig. 2a,b). While the 1 MHz system required 193 mW of average power at the sample surface to achieve a mean SNR_{base} of 2.2 at a depth of 820 µm, TrUDI reached the same imaging depth with only 7.5 mW of average power while achieving a mean SNR_{base} of ~6.1. This is consistent with our numerical estimates that TrUDI's lower repetition rate leads to a ~250-fold increase in 3p excitation efficiency.

In addition, to eliminate potentially confounding differences between microscopes and enable a direct comparison within the same apparatus, animal and three-dimensional (3D) FOV, we mimicked the reference 1-MHz imaging conditions on the TrUDI optical path. As operating TrUDI's laser system in the MHz regime was not possible, we used its 160-kHz laser to spatially oversample the FOV by slowing down the scanner, effectively emulating a 1.3 MHz system. The use of the same pulse energy in both imaging modalities led to an eightfold higher 'extrapolated' average power (P_{extrap}) for the 1.3-MHz configuration compared to the actual average power (P_{actual}) used in TrUDI. Moreover, to keep the pixel pitch (1.8 µm) constant between the two modalities, we averaged the signal from eight successive pulses into a single pixel. Using this approach to mimic the reference approach, we compared it to TrUDI under two different conditions: one in which the pulse energy was kept the same (Fig. 2c versus Fig. 2d), and another in which the average power (P_{actual} and P_{extrap} for TrUDI and the 1.3-MHz configuration, respectively) was kept the same in both systems (Fig. 2e versus Fig. 2f).

In the first scenario, we performed imaging in the hippocampus through intact cortex at a depth of 1,080 µm using TrUDI and found that only ~18 mW of average power (P_{actual}) at the sample surface was sufficient to resolve neurons with estimated mean SNR_{base} of ~2.2 (Fig. 2c). This corresponds to a pulse energy of ~ 3.1 nJ at the focus, similar to the energy used in other 3p imaging studies[36,42,44,62]. In contrast, imaging

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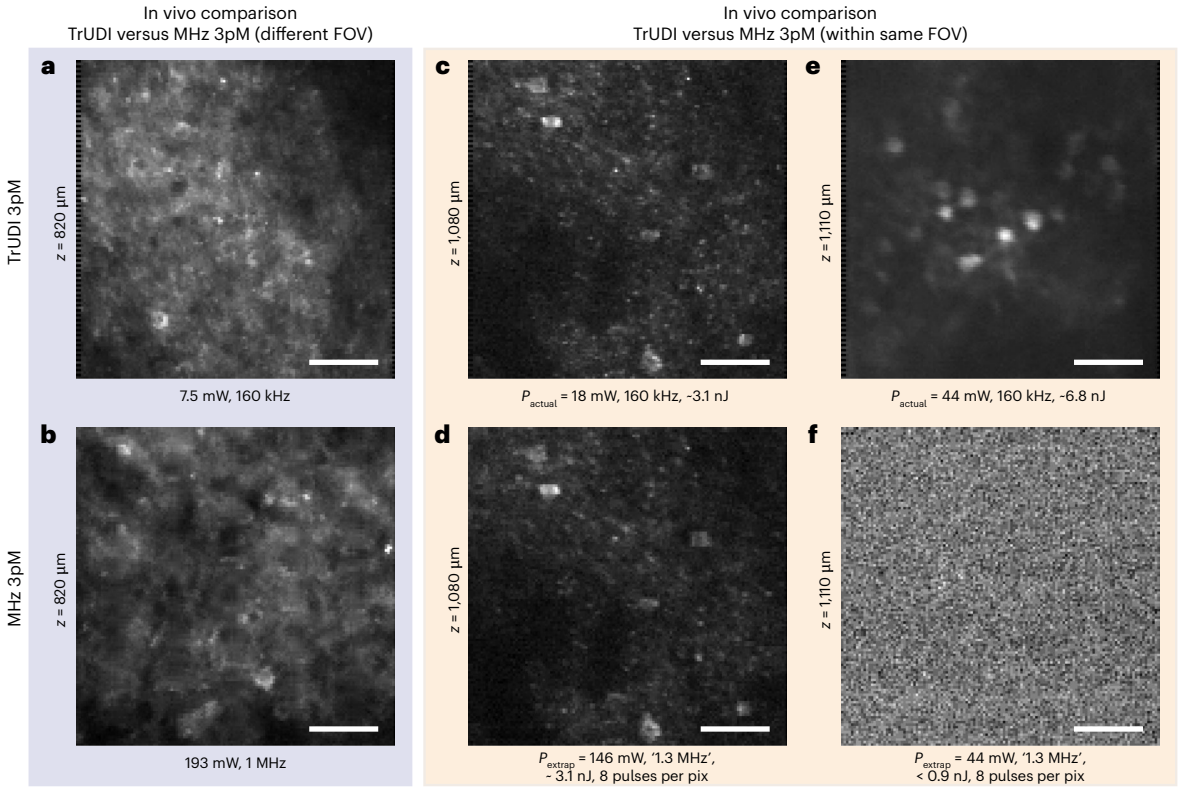


Fig. 2 | Comparison of in vivo three-photon microscopy (3pM): TrUDI versus typical 1-MHz-laser-based 3pM. a,**b**, Comparison of in vivo imaging between TrUDI and a 1-MHz-laser-based 3pM to reach an imaging depth of 820 µm within the PPC in mouse expressing GCaMP6s: TrUDI's 160 kHz system required an average power of ~7.5 mW (~3 nJ at the focus) (**a**) and 1 MHz system required 193 mW (**b**) (TrUDI had a pulse duration of ~50 fs, whereas that of the 1-MHz system was ~100 fs.). Scale bars, 50 µm. **c**,**d**, Comparison of the required average power between TrUDI and 1-MHz 3pM configuration for a fixed pulse energy within the same imaging FOV for similar pixel SNR at 1,080 µm depth in HPC of a mouse expressing GCaMP8m. TrUDI required an average power (P_{actual}) of 18 mW (~3.1 nJ at the focus) (**c**) and the 1-MHz configuration required an average power (P_{extrap}) of ~146 mW for the same pulse energy at the focus (**d**). Scale bars, 50 µm. **e**,**f**, Comparison of the SNR between TrUDI and 1-MHz 3pM configuration using the same average power within the same imaging FOV at 1,110 µm depth in HPC of a mouse expressing GCaMP8m. TrUDI with its 160-kHz repetition rate yields resolvable neurons with a mean SNR_{base} of ~4.8 using an average power (P_{actual}) of ~44 mW (**e**), whereas the SNR_{base} for the 1-MHz configuration using the same extrapolated average power (P_{extrap} = 44 mW) is too low to resolve neurons (**f**). Scale bars, 50 µm.

at the same depth with the same pulse energy in the 1.3-MHz configuration required ~146 mW of average power (P_{extrap}) (Fig. 2d). Thus, MHz-rate 3pM systems would well exceed the safe exposure limit at the SBR-limited imaging depth (>2 mm).

In the second scenario, we used ~44 mW of average power (P_{actual}) at the surface in TrUDI, which resulted in clearly resolved individual neurons at 1.1-mm depth (mean SNR_{base} = ~4.8) (Fig. 2e). To maintain the same average power in the 1.3-MHz configuration (P_{extrap} = 44 mW), the pulse energy needed to be reduced eightfold to <0.9 nJ, which resulted in a ~500-fold lower 3p signal and thus prohibitively low SNR_{base} to identify and segment neurons (Fig. 2f).

These results demonstrate the potential of TrUDI for extending the obtainable imaging depths in the scattering mouse brain while requiring minimal average power. This is achieved by combining a lower-repetition-rate laser with a 3p acquisition system optimized to maximize in-focus fluorescence signal generation per pulse. We next explored the maximum in vivo imaging depth achievable with TrUDI and its ability to record neuronal population activity at unprecedented tissue depths in the mouse brain.

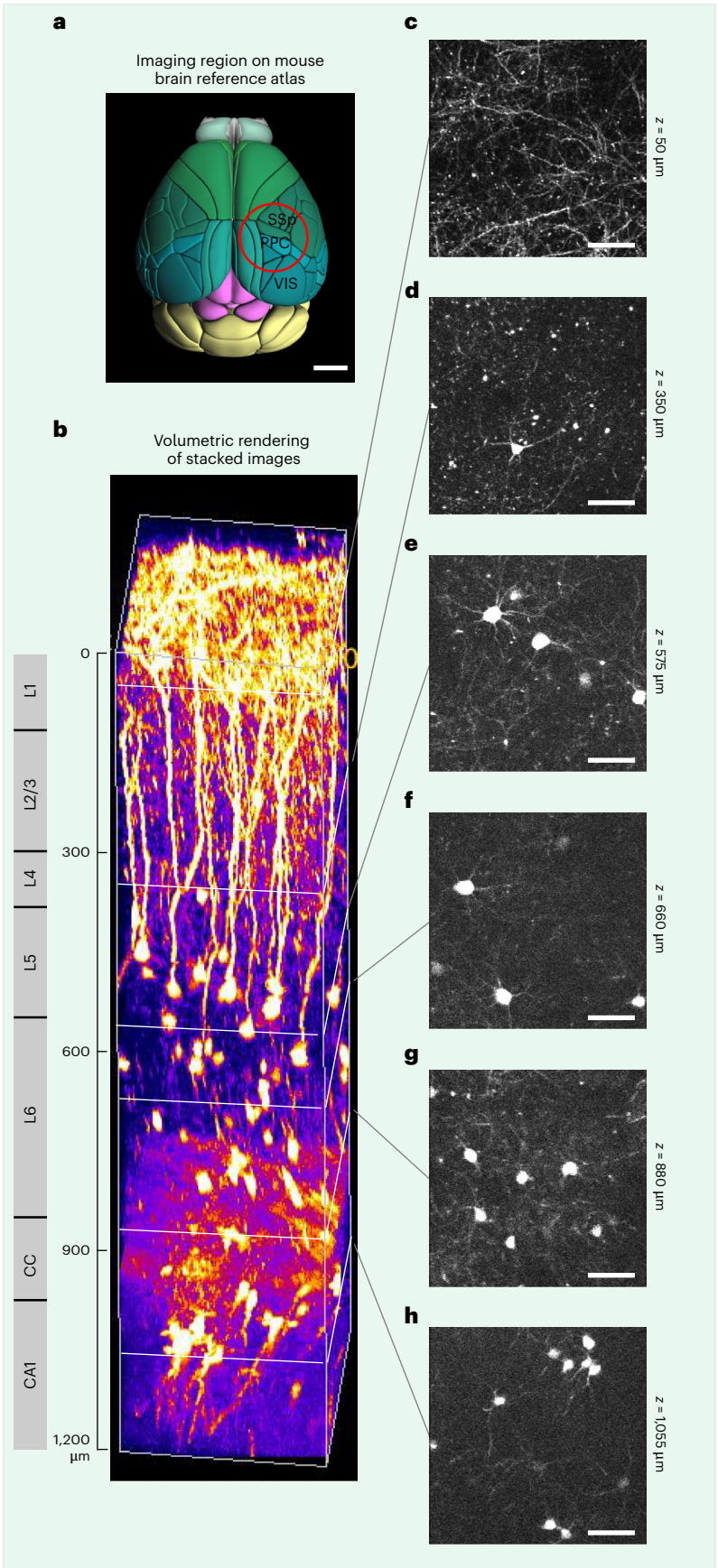
Structural imaging and neuronal population recording in the mouse hippocampus through intact cortex at 1,730 µm depth

Based on the above gains in in-focus fluorescence signal, we expected that TrUDI would enable both, higher-SNR imaging at the previously demonstrated tissue depths at lower average power, and imaging at previously unattainable tissue depths within the safe average power limits. To demonstrate this, we imaged through the cortical layers in the PPC of mice expressing green fluorescent protein (Thy1-EGFP) into the HPC region (Fig. 3a,b). We obtained a high-resolution reconstruction of the acquisition volume using an axial plane spacing of 5 µm and extending to 1,200 µm depth (Fig. 3b), along with selected single planes (Fig. 3c–h). Detailed neuronal processes were observed throughout the PPC and the dorsal HPC, using ~76 mW at the deepest imaging plane at ~1,200 µm, beyond the corpus callosum (CC). Additionally, we found highly detailed neurite structures associated with the CA1 pyramidal layer within a depth range of 1,050–1,250 µm (Fig. 3i–m). But more notably at depths approaching the SBR limit, we could clearly resolve individual neurons with high SNR_{base} (>7.0) at depths up to 1.73 mm, using ~105 mW of average power, well below established thresholds for thermal damage[36,42,44,63] in multiphoton microscopy (Fig. 3n).

Neuronal population imaging in posterior parietal, primary visual cortex and hippocampus at up to 1,840 µm depth

To demonstrate capturing of neuronal population activity at large tissue depth, we imaged transgenic mice expressing GCaMP8s or GCaMP8m throughout the cortex and subcortical regions. To do so, we captured the activity of neurons within a FOV of 230 × 230 µm^2 through various layers of the PPC or the primary visual cortex (V1), and further into the hippocampus, in head-fixed mice during spontaneous activity (Fig. 4a,b) at 8.6 Hz. This allowed us to obtain 3D representations of

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Fig. 3 | TrUDI enables deep structural imaging of the HPC through intact PPC in mice expressing green fluorescence protein (Thy1-EGFP). a, Rendering of the mouse brain based on the Allen mouse brain atlas[73] showing the approximate location of a 4-mm cranial window (red circle) over the PPC and the subcortical HPC, from which the recordings in **b**–**n** were obtained. Two neighboring areas, the primary somatosensory cortex (SSp) and the visual cortex (VIS) are also marked. Scale bar, 2 mm. **b**, High-resolution static volume stack covering ~1,200-µm depth range showing detailed neuronal processes in different layers (L1–L6) in PPC and CA1 layer of HPC through the CC, acquired using between 0.7 and 76 mW of average power, with 76 mW being used for the deepest layer of CA1 in HPC. Mean intensity projection over 20 frames per plane was used. Size of the imaged volume: 230 × 230 × 1,200 µm^3 using 0.9 × 0.9 × 5 µm^3 pixel spacing. **c**–**h**, Single-plane slices of the volume in **b** are shown for selected depths at 50 (**c**), 350 (**d**), 575 (**e**), 660 (**f**), 880 (**g**) and 1,055 (**h**) µm, with corresponding average acquisition power values of 0.8, 1.3, 2.0, 3.0, 14.0 and 54.0 mW, respectively. Scale bars, 50 µm. **i**,**j**, Two 3D-rendering views (top skew view (**i**) and side skew view (**j**)) of high-resolution recordings of the dorsal hippocampus covering a depth range of 1,050–1,250 µm generated using an average power of 50 mW. Size of imaged volume was 230 × 230 × 200 µm^3 using 0.9 × 0.9 × 5 µm^3 pixel spacing. **k**–**m**, Single-plane slices corresponding to depths at 1,150 (**k**), 1,185 (**l**) and 1,225 (**m**) µm, respectively, within the deeper range of volume shown in **j**, revealing detailed views of dendrites, soma and stems of axons. Scale bars, 50 µm. **n**, An example single-plane recording at 1,730 µm, a depth about ~2 EAL greater than previously achievable limits[36,43], acquired using 105 mW of average power, which is substantially lower than the established upper bound for safe power limits. Mean intensity projection over 20 frames. Size of imaged area was 230 × 230 µm^2 using 1.8 × 1.8-µm pixel spacing. Scale bar, 50 µm.

time-projected recordings across 170 planes, axially spaced by 10 µm reaching a depth of ~1.7 mm while using only 127 mW, a substantial part of which was needed to pass through the CC, due to its strong scattering properties (Fig. 4b and Supplementary Video 1). For comparison, the required power for imaging at 750 µm was only ~ 4.7 mW, whereas imaging at 1,030 µm required an average power of ~21 mW.

We performed 6–10-min recordings at 8.6 Hz across multiple planes including deeper cortical layers in PPC, hippocampal CA1 and DG neuronal populations ranging from 730 to 1,840 µm (Fig. 4c–g). We segmented neurons and extracted the time traces of their activity in each plane after performing motion correction (Extended Data Fig. 6), using a combination of spatiotemporal filters and non-negative matrix factorization implemented via our custom CaImAn-based[64] pipeline.

Within each plane, neurons could be clearly identified and segmented, and their Ca^{2+} dynamics reliably extracted across the full depth range. We show example neurons and time traces from recordings within a depth range of 1,040–1,840 µm (Fig. 4h–l), along with additional active neurons and their corresponding time traces in each plane (Extended Data Fig. 7 and Supplementary Video 2). The maximum average power in these recordings was ~66 mW at the deepest depth of 1,840 µm substantially below that for Thy1-GFP (Fig. 3n) and the GCaMP8m (Fig. 4b) mice at comparable depths potentially due to the higher brightness and sensitivity of the tetO-GCaMP8s indicator under the Camk2a-tTA promoter.

TrUDI enables neuronal population imaging through motor cortex and the striatum at up to 1,930 µm

Next, we validated the performance of TrUDI by recording neuronal population activity across depths and layers of the secondary motor cortex (M2) and the striatum (Fig. 5a) in GCaMP6s expressing mice during spontaneous activity. Using a 20-µm axial plane spacing and otherwise the same imaging parameters as in Fig. 4b, we recorded the activity of neurons from the superficial layers of the motor cortex down to ~2.1 mm depth from the pia. We present the reconstructed volume of time-projected neuronal population activity, where active neurons are clearly visible across different depths while using ~150 mW maximally for our deepest recordings at ~2.1 mm depth (Fig. 5b). We show time-projected images from 4–10-min-long recordings of individual planes at 8.6 Hz, within a depth range of 1.05–1.93 mm using the same data processing pipeline as in Fig. 4 (Fig. 5c–g), along with time series of indicated neurons displaying clear Ca^{2+} transients through depths up to 1,930 µm (Fig. 5h–l). This capability is further illustrated in another dataset where depth resolved activity pattern of individual neurons across planes is shown while using a maximum power of ~120 mW for the deepest plane at 1,930 µm (Extended Data Fig. 8 and Supplementary Video 3).

Our results clearly demonstrate the capabilities of our new imaging system. By performing systematic and careful optimization of excitation laser parameters, acquisition modality and the optical parameters of our 3pM system, we demonstrated a ~250-fold effective increase in in-focus fluorescence signal within the established safe limits of average power and energy density. This enabled Ca^{2+} imaging across ~6 EAL. Our demonstrated imaging depth of ~2 mm through intact brain tissue exceeds previous 3pM achievements by ~800 µm (corresponding to ~2–3 EALs) and approaches the fundamental SBR-limited imaging depth in 3pM at 1,300 nm in the mouse brain.

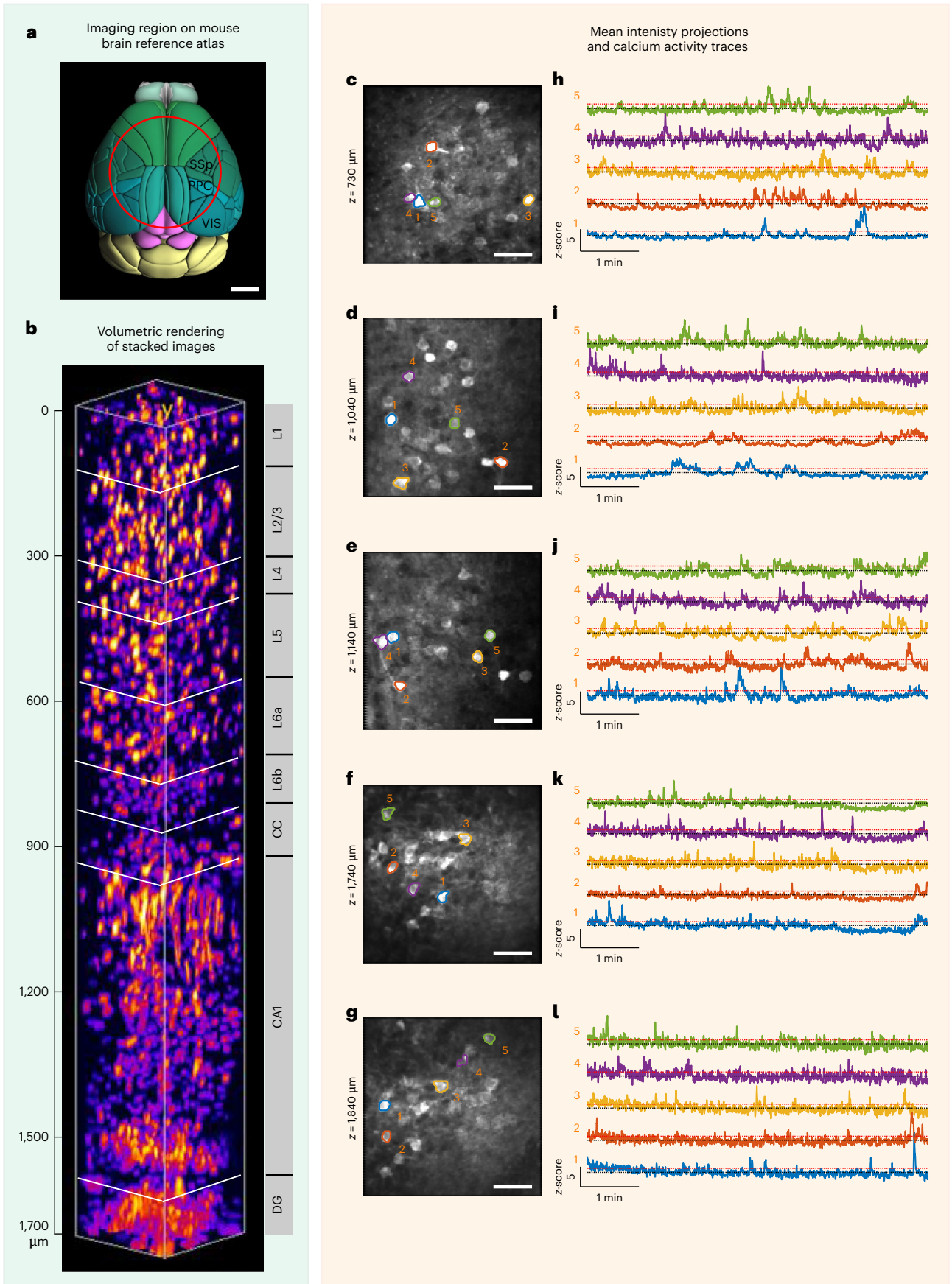
Immunohistochemical validation of safe limits for linear and nonlinear laser exposure

While the use of a maximal average power level <150 mW out of the microscope objective and a peak fluence of ~0.3 nJ µm^{-2} for our most demanding condition during imaging at ~2-mm tissue depth were informed by previous experimental studies on the safe limits of linear and nonlinear laser exposure of the sample[42], we set out to investigate and validate these at the specific parameter regime used by TrUDI.

We used GFAP, a reporter of astrocytic activation, NeuroTrace as a fluorescent Nissl stain labeling neuronal somata as a reporter of neuronal morphology and structural integrity, and γH2AX as a marker of DNA damage. We utilized all markers while comparing their respective responses in our most extreme experimental conditions (imaging

Fig. 4 | TrUDI enables deep-tissue Ca^{2+} imaging over a ~1,800-µm depth range, allowing imaging of the majority of HPC layers through intact PPC in mice expressing GCaMP8m and GCaMP8s. a, Rendering of the mouse brain based on the Allen mouse brain atlas[73] showing the approximate location of an 8-mm cranial window (red circle) over the PPC and the subcortical HPC, from which the recordings in **b**–**l** were obtained. Scale bar, 2 mm. **b**, High-resolution reconstructed volume stack covering a ~1,700-µm depth range, showing varying density of neurons across different layers (L1–L6) of the PPC, as well as the neurons in the hippocampal CA1 and DG region of mouse expressing GCaMP8m (VGAT-Cre × GCaMP8m). The acquisition power was increased exponentially between 0.4 and 127 mW to obtain constant signal across depths. Mean intensity projection based on recording of 1-min duration at 8.6 Hz for each layer. The size of the imaged volume was 230 × 230 × 1,700 µm^3 using 1.8 × 1.8 × 10-µm^3 pixel spacing. **c**–**g**, Mean intensity projection of neuroactivity in individual planes at depths 730 (**c**), 1,040 (**d**), 1,140 (**e**), 1,740 (**f**), and 1,840 (**g**) µm, respectively, in mouse brain expressing GCaMP8s (Camk2a-tTA × tetO-GCaMP8s), showing resolvable neurons and corresponding Ca^{2+} activity traces (**h**–**l**), even in the deepest recordings at 1,840 µm. The average power used for each of the five depths was 2.5, 6.5, 12, 58 and 66 mW, respectively, all of which were substantially lower than the established upper bound for safe power limits. Imaging area, 230 × 230 µm^2; pixel size of 1.8 × 1.8 µm^2. Scale bars, 50 µm. **h**–**l**, Ca^{2+} activity traces of selected neurons numbered in **c**–**g** for depths 730 (**h**), 1,040 (**i**), 1,140 (**j**), 1,740 (**k**) and 1,840 (**l**) µm, respectively, over ~6 min. Traces were extracted from recordings using CaImAn[64] with *z*-scores up to ~10 for the strongest responding neurons. The red dotted line denotes a *z*-score of 1 and the black dotted line denotes the baseline.

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a FOV of ~230 × 230 μm^2 at 2 mm at 8.6 Hz using 150 mW for imaging over a period of 10 min), to those under negative and positive control conditions. Additionally, we explored the limits of a potential 'head room' for laser exposure by comparing the responses of the above markers in samples exposed to average power levels of 200 mW at otherwise equal conditions to our most demanding experimental condition (Extended Data Fig. 9).

We found no significant increase in the GFAP fluorescence response ($P = 0.39$, one-sided Welch's t-test) due to astrocyte activation under our most extreme experimental condition compared to the negative control (Extended Data Fig. 9a); however, using 250 mW at 2-mm depth and otherwise equal conditions as in our experimental conditions did result in a ~2× increase in GFAP response ($P = 0.002$, one-sided Welch's t-test) within a 1 × 1 mm^2 area located within the cortex overlying the CC (Extended Data Fig. 9d). At 200 mW we observed a moderately elevated (~1.5×) immunoreactivity response ($P = 0.014$, one-sided Welch's t-test) suggesting a borderline condition, which depending on other imaging parameters and experimental conditions might still be potentially viable.

Consistent with the above results using neuronal somata marker, NeuroTrace, we did not detect a reduction in labeled cell density ($P = 0.80$, one-sided Welch's t-test) as would be expected if laser exposure would have caused neuronal loss or structural damage (Extended Data Fig. 9b); however, under our positive control condition at 250 mW we did observe some degree of laser-induced cell loss (~50% lower cell density compared to negative control; $P = 0.004$) (Extended Data Fig. 9e). Unlike the results of GFAP response, however, at 200 mW the response of NeuroTrace was comparable to the negative control condition with no significant reduction of cell density ($P = 0.45$, one-sided Welch's t-test).

While the above results established that even our most demanding experimental condition using 150 mW at 2-mm tissue depth was within safe limits of laser exposure, we note that they deviated from observations by Wang et al.[42] where the first onset of a positive immune response was observed at 150 mW. This difference can be accounted for by considering their application of this power level at a substantially lower imaging depth of ~1 mm in addition to a number of other factors, including use of the slightly longer wavelength of 1,320 nm resulting in ~20% more absorption, a smaller cranial window size of 4 mm, as opposed to 8 mm in our case, which substantially facilitates heat dissipation.

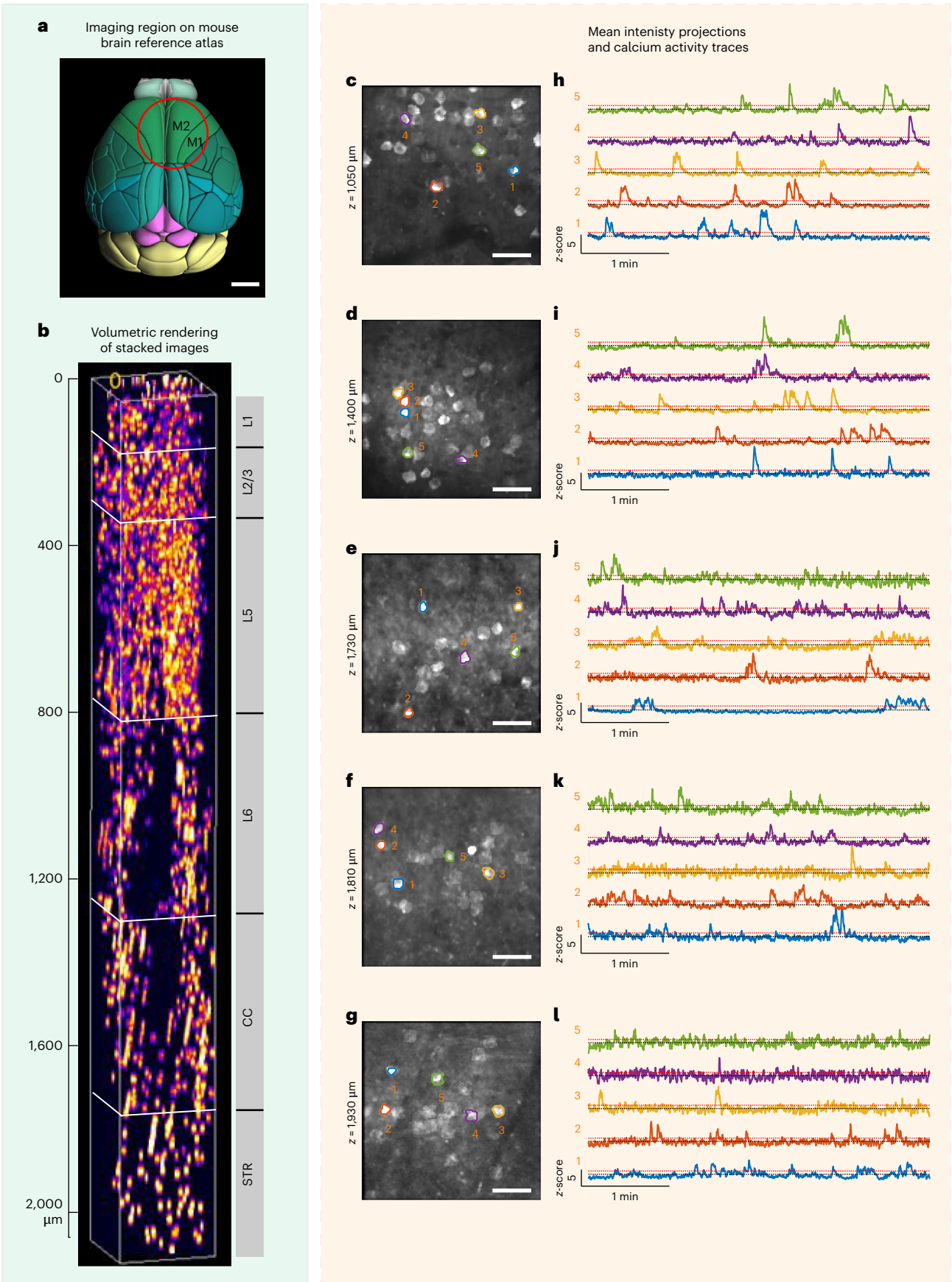
Thus, to corroborate our above experimental findings and rationalize how differences in imaging parameters and conditions at the same average power could account for the observed differences in the immunohistochemical responses, we performed experimentally informed simulations using a previously developed[63,65] and utilized[41,42,51] method. Using established numerical values for tissue scattering[42] and water absorption[66], the simulation obtains the fraction of absorbed photons within the tissue volume for a given imaging condition, including depth and scanning FOV, using a Monte-Carlo light transport model, which is followed by numerical recipe for diffusion of heat[65] to compute the temperature distribution within the tissue at various laser power levels (Supplementary Note 6 and Extended Data Fig. 10). Consistent with our above experimental results, we found that even under our most demanding experimental conditions using ~150 mW at 2-mm depth, only a peak temperature increase of ~0.3 °C (Extended Data Fig. 10b,g) was induced, well within established physiologically tolerable range[42,63,65]. Similarly for the 200 mW at 2-mm depth conditions, only a slightly higher peak temperature increase of ~3 °C was induced (Extended Data Fig. 10c,g) consistent with a slightly elevated GFAP response. This was in contrast to the 250 mW condition at 2-mm depth, for which we observed increased GFAP labeling alongside a reduction in NeuroTrace-labeled neuronal cell density. Under this condition a temperature increase of ~5.7 °C compared to baseline was observed (Extended Data Fig. 10d,g). To rationalize the observed difference in GFAP response in our experimental conditions and the condition used in Wang et al.[42] where 150 mW was used at 1-mm tissue depth, we ran the same simulations for the imaging parameters used in that work and could indeed observe a temperature increase of 4.4 °C (Extended Data Fig. 10f) consistent with their immunohistochemical and simulation results on the initial onset of immune response under these conditions.

To experimentally validate that our most demanding experimental conditions were also below limits of safe laser exposure with regards to nonlinear and free-radical-mediated damage mechanisms we investigated the response of our samples labeled with γH2AX (Extended Data Fig. 9c,f,g). It has been suggested that the high irradiance of the laser used in multiphoton microscopy can generate free radical through interaction with water, which may in turn initiate DNA fragmentation for which γH2AX has been shown to represent an effective marker[67]. Using γH2AX-labeled samples at higher magnification we did not observe any noticeable increase ($P = 0.96$, one-sided Welch's t-test) in γH2AX fluorescence in our most demanding experiential condition at 150 mW at 2-mm depth corresponding to a peak fluence of ~0.3 nJ μm^{-2} compared to the negative control (Extended Data Fig. 9c). Similarly, no observable effects ($P = 0.65$, one-sided Welch's t-test) were detected at 200 mW and a peak fluence of ~0.4 nJ μm^{-2} under the same imaging conditions, suggesting that even this somewhat higher power level is likely within the safe limits of laser exposure with respect to nonlinear damage at our experimental parameters. While an increase in γH2AX fluorescence relative to the negative control ($P = 0.048$, one-sided Welch's t-test) was observed under the positive control condition at 250 mW and a peak fluence of ~0.5 nJ μm^{-2} (Extended Data Fig. 9f) when quantified over the unsegmented regions of interest similar to the manner used for GFAP and NeuroTrace analysis, even under this condition we did not observe intranuclear γH2AX accumulation or a significant change ($P = 0.92$, one-sided Welch's t-test) in mean fluorescence over the nuclear area (Extended Data Fig. 9g) or punctate foci, which are typically associated with DNA fragmentation[67].

Moreover, this increased γH2AX immunostaining was particularly pronounced in regions affected by cell loss, with a diffuse, pan-nuclear pattern associated with cellular breakdown and debris rather than discrete nuclear foci characteristic of a

Fig. 5 | TrUDI enables deep-tissue Ca^{2+} imaging over ~2,000-μm depth range within motor cortical areas and the striatum of mice expressing GCaMP6s. **a**, Approximate location of a 5-mm cranial window (red circle) over the mouse brain, targeting deep imaging in the secondary motor cortical area (M2) and striatum, shown over the Allen mouse brain atlas[73]. Scale bar, 2 mm. **b**, High-resolution reconstructed volume stack covering a ~2,060-μm depth range showing varying density of neurons across different layers (L1–L6) of M2 as well as the neurons in the striatum (STR) for mouse expressing GCaMP6s (Camk2a-tTA × tetO-GCaMP6s). The acquisition power was exponentially increased from 0.5 to 150 mW to obtain constant signal across depths. The size of the imaged volume was 230 × 230 × 2,060 μm^3 using 1.8 × 1.8 × 20 μm^3 pixel spacing. Images show mean intensity projections over ~1 min at 8.6 Hz. **c**–**g**, Mean intensity projection of neuroactivity in individual planes at depths 1,050 (**c**), 1,400 (**d**), 1,730 (**e**), 1,810 (**f**) and 1,930 (**g**) μm, respectively, in mouse brain expressing GCaMP6s, showing resolvable neurons and corresponding Ca^{2+} activity traces (**h**–**l**), even in the deepest recordings at 1,930 μm. The average power used for each of the five depths was 10, 30, 120, 120 and 120 mW, respectively, all substantially lower than the established upper bound for safe power limits. Imaging FOV, 230 × 230 μm^2; pixel size of 1.8 × 1.8 μm^2. Scale bars, 50 μm. **h**–**l**, Ca^{2+} activity traces of selected neurons numbered in **c**–**g** for depths 1,050 (**h**), 1,400 (**i**), 1,730 (**j**), 1,810 (**k**) and 1,930 (**l**) μm, respectively, over ~4 min. Traces were extracted from recordings using CaImAn[64] and show z-scores up to ~12 for the strongest responding neurons. The red dotted line denotes a z-score of 1 and the black dotted line denotes the baseline.

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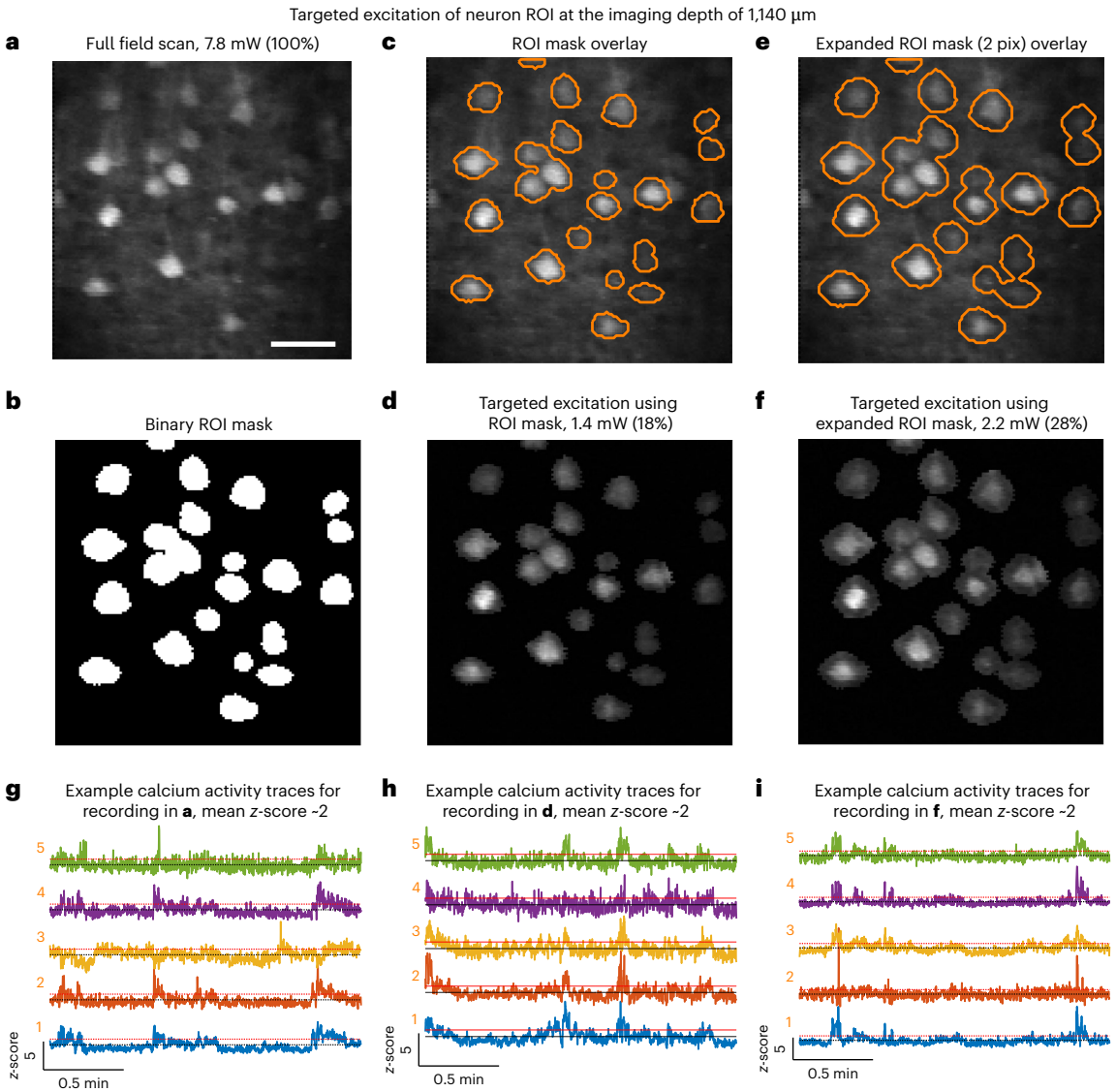


Fig. 6 | Targeted excitation minimizes laser exposure in TrUDI. a, A representative full-field scanned image at otherwise equal conditions used in this work acquired using an average power of 7.8 mW. Imaging depth, 1,140 µm. **b**, Extracted neuronal ROIs after spatial segmentation using a binary mask generated via Cellpose in MATLAB[74] for the image shown in **a**. **c**, Overlay of binary ROI mask extracted in **b** with image **a**, resulting in a pixel coverage of ~18% of the area of the full-field image. **d**, Image acquired with power modulation over the same FOV as in **a** using the ROI mask overlay shown in **c**, resulting in a total average power of 1.4 mW (~18% of that used for full-field imaging). **e**, Overlay of expanded ROI by 2 pixels along the perimeter of the ROIs extracted in **b** resulting in a pixel coverage ~28% of the area of full-field image. **f**, Image acquired with power modulation over the same FOV as in **a** using the expanded ROI mask overlay shown in **e** resulting in a total average power of 2.2 mW (~28% of that used for full-field imaging). **g**–**i**, Ca^{2+} activity traces of representative neurons for the recording in **a**, **d** and **f** are shown in **g**, **h** and **i**, respectively. Traces were extracted from recordings using CaImAn[64] after motion correction. The mean *z*-score in all three recordings is ~2. The red dotted line denotes a *z*-score of 1 and the black dotted line denotes the baseline. For all panels, image size, 230 × 230 µm^2 with 1.8 × 1.8 µm^2 pixel spacing. Images show mean intensity projections over ~2–6 min at 8.6 Hz. Scale bar, 50 µm.

localized DNA damage response[67–69]. This suggests that while a direct peak-fluence-induced DNA fragmentation or nonlinear damage mediated by free radical formation is likely not occurring even at the 250 mW condition at 2-mm depth, there are other mechanisms in place, such as localized heating, which can cause cell damage resulting in histone accumulation as marked by γH2AX. Thus, although the exact mechanism resulting in the observed γH2AX response remains to be determined, our results indicate that the 250-mW condition at 2-mm depth exceeds the limit for safe laser exposure for long-term imaging, whereas even our most demanding exposure conditions in this work with 150 mW at ~2-mm tissue depth and a peak fluence of ~0.3 nJ µm^{-2} combined with our other imaging conditions fall within the limits of both, safe linear and nonlinear laser exposure (Fig. 1g,h).

Targeted excitation in TrUDI allows for up to 80% further reduction of laser power exposure

To account for possible imaging scenarios such as deeper imaging, imaging in more scattering tissues or to relax radiation stress during long-term and repeated imaging we devised a selective excitation strategy that allowed for a substantial reduction of the required excitation power.

We found that even for densely labeled samples typically ~50–80% of pixels in the imaging FOV did not overlap with any neuronal cell bodies or provided sufficient SNR despite of the above optimization of energy density (Fig. 6a,b). Thus, guided by a short (~2 min) acquired mean intensity projection image that allowed us to identify the image pixels with sufficient overlap with neurons and using a fast electro-optical modulator (EOM) allowing for binary pulse-by-pulse

modulation of excitation, we selectively blocked the sample excitation of those pulses that did not contribute to neuronal signal generation. This was carried out by first spatially segmenting neurons identified in our short mean intensity projection recording and subsequently defining region of interest (ROI) masks (Fig. 6c,e) that included marginal pixels that provided a 'buffer' for animal motion. This mask was then used as a template for a binary power modulation via the EOM during scanning of the FOV. As an example, we demonstrated that using this approach at 1,140 µm depth the average laser power can be reduced to ~18–28% (Fig. 6d,f) compared to the conventional full FOV illumination approach while maintaining the same mean *z*-score of ~2 in the Ca^{2+} activity traces (Fig. 6g–i). We note that a conceptually similar approach in which laser pulses during the amplification process are modulated has been previously also shown[70].

Discussion

In the last decade, 3pM has emerged as a powerful microscopy technique for neuroscience by extending the imaging depth reach beyond the ~600 µm possible with 2pM in the scattering mammalian brain; however, its key limitation is the unproportionally high heat penalty due to the higher-order nonlinearity compared to 2pM. Thus, all demonstrations of 3pM to date have been constrained by the in-focus signal achievable within safe average power limits. In contrast, our results show that without this limitation, the true SBR-limited depth reach of 3pM at 1,300 nm (the optimal wavelength for exciting green Ca^{2+} indicators such as GCaMP variants) is ~2.3 mm. The fact that previous methods have failed reaching this depth reflects partially ad hoc choice of certain imaging and instrument parameters and lack of systematic optimization. Here we demonstrate how the fundamental SBR-limited imaging depth in 3pM can be achieved through a bottom-up principled optimization of the optical and acquisition parameters of a 3p microscope, particularly the excitation repetition rate, PSF, pulse duration and pulse energy, while maintaining the required spatiotemporal resolution and remaining within the linear and nonlinear constraints of laser exposure.

While we have shown that reaching the SBR-limited depth in 3pM is possible without any adaptative optics (AO) strategies[53,54,71,72], the excitation light undergoes major aberrations as it traverses through the refractive index inhomogeneities in tissue. Thus, we expect that in the future, a combination of our method with suitable AO techniques would allow for correction of the PSF aberrations and further enhance in-focus fluorescence, enabling higher-SNR recordings and resolution of finer features at large tissue depths. We do not expect AO methods to enable major increases in imaging depth, as the demonstrated AO signal improvements of ~5–10 times[53,71,72] translate only to an imaging depth increase of ~150–250 µm; however, beyond enabling finer feature resolution, AO could also further reduce average power requirements. A similar argument can be made for the further optimizations of the detection path. A combination of custom high-NA, long-working-distance objectives with improved transmission efficiency at the excitation and detection wavelengths and larger aperture optics in the detection path can improve fluorescence photon collection efficiency, which scales quadratically with the effective collection angle, thereby reducing power requirements.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41592-026-03251-2.

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Methods

Experimental model and animal preparation

All surgical and experimental procedures were approved by the Institutional Animal Care and Use Committee of The Rockefeller University. Mice were obtained from The Jackson Laboratory and bred in-house for the specified transgenic lines. Mice were housed in pairs on a reverse dark–light cycle (08:00 to 20:00 dark and 20:00 to 8:00 light) at normal room temperature and humidity. In this study, we used mice of 4–18 weeks of age and expressing green fluorescence protein (Thy1-EGFP, JAX 007788) for static imaging experiments, as well as transgenic lines Camk2a-tTA (JAX 007004) × tetO-GCaMP6s (JAX 024742), Camk2a-tTA (JAX 016198) × tetO-GCaMP8s (JAX 037717) and VGAT-Cre (JAX 016198) × GCaMP8m (JAX 037718) for Ca^{2+} activity imaging. For experiments involving the study of linear and nonlinear laser-induced adviser effect, we used the wild-type mouse, C57BL6J (JAX 000664).

Cranial window surgery

Cranial window implantation was performed under isoflurane anesthesia (1–1.5% maintenance at a flow rate of 0.7–0.9 l min^{-1}) using procedures adapted from previous studies[41,51,75]. After mice were placed in a stereotaxic frame (RWD Life Science), dexamethasone (0.4 mg ml^{-1}, subcutaneously) and meloxicam (0.125 mg kg^{-1}, subcutaneously) were administered preoperatively to reduce brain swelling, suppress inflammatory responses and stabilize the blood–brain barrier. Additionally, bupivacaine (0.125 mg ml^{-1}) was injected subcutaneously above the skull to provide local anesthesia and protect tissue from excessive neuronal excitation. The scalp was sterilized with betadine and ethanol, then removed via midline incision to expose the underlying skull. The periosteum and connective tissue were carefully cleared using a scalpel and fine forceps. A custom-made stainless-steel head bar was fixed posterior to the occipital bone using cyanoacrylate glue (Loctite) and reinforced with black dental cement (Ortho-Jet, Lang Dental). A craniotomy was performed using a high-speed dental drill (0.5-mm burr) with continuous irrigation to prevent overheating. The size and shape of the craniotomy varied depending on the imaging requirements, ranging from circular (3-mm diameter) for single-hemisphere imaging to D-shaped (7–8 mm) for enhanced bilateral cortical optical access. A single or two bonded sterile glass coverslips (no. 1 thickness, Warner Instruments; optical adhesive Norland 60) were implanted into the craniotomy and sealed in place using tissue adhesive (Vetbond).

Following surgery, mice received dexamethasone and meloxicam (as previously described) for 7 consecutive days, to minimize immune activation and provide analgesia. Additionally, antibiotic-enriched feed (Sulfatrim) was provided to prevent infection and support physical recovery. Mice were housed in their home cages under standard conditions and allowed to recover for at least 2–3 weeks before imaging. Animals were monitored daily for signs of infection or discomfort and were excluded from imaging experiments upon detection of dura damage or compromised brain tissue clarity.

TrUDI microscope for 3p imaging

A schematic of the optical path of the 3p microscope is shown in Extended Data Fig. 5. To meet the pulse energy requirements as well as the optimal repetition rate across the near-infrared and infrared wavelength ranges, we used a factory-customized tunable OPA system. The laser system comprised an ultrafast diode-pumped ytterbium-doped regenerative amplifier (Yb:KGW, ytterbium-doped potassium gadolinium tungstate, Carbide CB3, Light-Conversion, 80 W, 500 µJ output at 1,030 nm, ~210 fs, 160 kHz) as a pump for the hybrid OPA (Orpheus-F, short pulse mode, Light-Conversion). A part of the pump pulse energy was used to generate a white light continuum by focusing it onto a nonlinear crystal and the rest was used to generate second harmonics required for pumping the OPA stages at 515 nm. The white light was used as the seed for the OPAs by tuning the delay and chirp to select the spectrum and bandwidth for the 3pM wavelength. The weak seed was then amplified in two stages using the second harmonic pump at the OPA crystals in a hybrid non-collinear and collinear fashion. The idler wave, required for 3p imaging, was then separated from the seed wavelength outputs, and dispersion was pre-compensated to achieve the shortest pulse duration based on the requirement for the downstream optical components. We tuned the OPA to our wavelength of interest, maximizing output power and bandwidth at 1,300 nm, resulting in a pulse energy >10 µJ with ~50 fs compressed pulse duration at the laser output at 160 kHz repetition rate.

For pulse compression and dispersion compensation for the downstream optical components, we used a prism compressor using a pair of Brewster-angle prisms (SF10) placed on an adjustable rail to optimize the pulse duration for 1,300 nm. The final pulse duration was optimized at the sample location by collimating the light out of the objective lens followed by free-space coupling to an autocorrelator (APE pulseCheck). At 1,300 nm, the minimum pulse duration obtained was approximately 50 fs with a Gaussian fit to the intensity autocorrelation, representing a full bandwidth over 85 nm. For power modulation, we used a fast EOM (Conoptics, 360-40) that handled both primary power adjustment and beam blanking at the turning edge of the fast-scanning axis; however, for imaging across a ~6–7 EAL depth range, the dynamic range for power modulation exceeded the contrast provided by the EOM. Therefore, we incorporated a secondary power adjustment module consisting of a motorized half-waveplate and a polarized beamsplitter (Thorlabs). Beam sizes at various locations were adjusted with telescopes consisting of off-the-shelf lenses (Thorlabs).

The laser output was interfaced with the 3p microscope (Sutter 3p-MOM) comprising a three-axis micrometer stage, galvo–galvo scanners and scan optics. We used a ×16 objective with a long working distance of 3 mm and NA 0.8 (Nikon, CFI75 LWD 16X W), unless otherwise noted. The detection assembly, which was based on Sutter MOM and was developed at the HHMI Janelia Research Campus as part of the MIMMS microscope had customed optics for 3pM, which included a primary dichroic mirror (Semrock, FF735-Di02, 25 × 36 mm) with a cutoff wavelength of 735 nm and cutoff width 30 nm for separating excitation (1,300 nm) and emission light (350–720 nm), an emission dichroic with cutoff centered at 565 nm and cutoff width ~15 nm (Chroma T565lprx 35.5 × 50 mm) for separating green and red signals, an emission fluorescence filter centered at 525 nm and a width of 70 nm (ET525/70m-2p for green, 30-mm diameter). For the collection of light, the emission reflected from the dichroic was coupled to the PMT (Hamamatsu, H10770-PA40, active area 5 × 5 mm) via an achromatic doublet (Edmund Optics, NT49-284-INK, f = 100 mm, 50-mm diameter) and aspheric condenser (Newport, KPA034-C, 28.5 mm EFL, 32.5-mm diameter) lenses. This arrangement allowed for collection and relaying of diverging rays emitted maximally up to ~8° from the collimated configuration from the objective back-focal-aperture to be mapped onto the active area of the PMT (Extended Data Fig. 2). Data acquisition and microscope control were handled with a single data-acquisition card (vDAQ, MBF) and ScanImage software (v.2022, MBF). The PMT signal was recorded with a 16-bit digitizer at 120 MS s^{-1} sampling rate. A clock-generation board (Texas Instruments) provided the necessary clock signals derived from the laser repetition rate for synchronized and gated sampling in a single-pulse-per-pixel acquisition scheme. The gate width was optimized to integrate the fluorescence signal for the best SNR.

Imaging protocol

Awake mice were head-fixed on a treadmill platform, and the flat glass window was aligned perpendicularly to the optical axis using a tip-tilt stage and an alignment tool based on a red laser diode and beamsplitter for generating a reference reflection. In a typical recording the image pixel resolution was 1.8 µm with an image size of 128 × 128 pixels, resulting in a FOV of 230 × 230 µm. In ScanImage, we set the temporal fill fraction of the fast axis to 0.9. With the single-pulse-per-pixel imaging and a 160-kHz laser, this yielded a frame rate of 8.6 Hz for recording

neuronal activity from a single plane. For high-resolution recordings, we used a pixel resolution of 0.9 µm and an image size of 256 × 256 while maintaining the same FOV. For *z*-stack recordings, the axial resolution was 5–20 µm, as noted in the main text and figure legends for individual cases. Depending on the recording depth, laser power to the EOM was adjusted using the secondary power controller. The EOM adjusted the laser power during changes of depth in single-plane and stack recordings and performed beam blanking during galvo mirror turnaround. The overall range of laser power used was ~0.5–150 mW, measured after the objective lens with a thermal power sensor (Thorlabs S405C) before each imaging session. As water has substantial absorption at 1,300 nm, fresh D_2O was used as the immersion medium for all recordings.

Experimental comparison of TrUDI with MHz 3pM

To compare the performance of a MHz imaging system with the optimized TrUDI microscope, we modified the sampling scheme of our 160 kHz system to mimic MHz conditions, enabling a more direct comparison between the two modalities as described in the main text. In brief, this was achieved by slowing down the scanner to spatially oversample the FOV, matching the sampling of an equivalent 1.3-MHz system, and subsequently averaging the signal from eight successive pulses into a single pixel to maintain a constant pixel size between configurations. This resulted in an eightfold higher 'extrapolated' average power (P_{extrap}) in the 1.3-MHz configuration compared to the actual average power (P_{actual}) used. Using this approach, we performed comparative in vivo imaging of a mouse expressing GCaMP8m in the hippocampus through the intact cortex. We compared TrUDI with the mimicked 1.3-MHz configuration under two conditions. In the first, the pulse energy at the focus was kept the same (~3.1 nJ) for both configurations while imaging at 1,080 µm. In the second, the average power was kept the same for both configurations (~44 mW; P_{actual} = 44 mW for TrUDI and P_{extrap} = 44 mW for the mimicked 1.3 MHz configuration) while imaging at a depth of ~1.1 mm.

Finally, to compare the imaging of TrUDI and a real 1 MHz system in mouse brain at a comparable depth, we constructed a secondary 3p setup following the design in one of our previous publications[36]. In brief, the 1 MHz system used an OPCPA laser (White Dwarf, Class 5) pumped by a fiber laser (80 W, 1,030 nm, 1 MHz, Coherent Monaco) to generate 1,300 nm light. The laser was then coupled to a Sutter microscope after passing through a power adjustment module, a pulse dispersion compensation module, and a scanner module consisting of a relayed resonance–galvo–galvo scanner configuration, followed by a scan lens (f = 110 mm, Thorlabs LSM05) and a tube lens (f = 400 mm, Edmund Optics), and the same objective lens used in TrUDI. The detection system was similar to the TrUDI setup. The data-acquisition system consisted of a high-speed FPGA-digitizer assembly (NI 7975R, NI 5772-02, PXI 6341, National Instruments) and ScanImage software (v.2018, MBF). Multi-ROI scanning with a 16-kHz resonant scanner and the galvo–galvo pair was configured to match the pixel resolution, achieve a similar FOV size, and provide an imaging speed of 30 Hz. The system was calibrated to have imaging parameters as close as possible to TrUDI, except for the pulse duration, which for OPCPA source was limited to ~100 fs (a factor of two longer than in TrUDI). We performed comparative in vivo imaging of deep cortical areas in the PPC of mice expressing GCaMP6s (Camk2a-tTA× tetO-GCaMP6s) on both microscopes (at nearly identical locations) (Fig. 2a,b). In TrUDI, we used 7.5 mW average power (~3-nJ pulse energy at the focus), whereas for the 1 MHz microscope, we used 193 mW power. Note that for all pulse energy calculations under the experimental conditions described here or elsewhere, we assumed a depth-dependent exponential loss using an EAL of 300 µm and ignored losses due to the immersion medium or sample surfaces.

Targeted excitation scheme to reduce sample exposure

As discussed in the main text, the targeted excitation scheme requires a template mask corresponding to neuronal pixels. To generate this target mask, conventional full-field illumination imaging was first performed at the desired depth using typical imaging parameters (FOV of 230 × 230 µm, image size of 128 × 128 pixels, pixel resolution was 1.8 µm, at a frame rate of 8.6 Hz) (Fig. 6a). The mean projection of the acquired images was used for spatial segmentation of neurons using Cellpose in MATLAB[74]. The resulting neuronal footprints were then converted into binary masks (Fig. 6b). Because our TrUDI microscope employs a one-pulse-per-pixel excitation scheme and utilizes a high-speed EOM capable of switching individual pulses on and off, the binary mask was directly fed into ScanImage to modulate excitation power across the FOV. The phase of the beam control signal, relative to the scanning galvo position, was optimized to align with the segmented mask, ensuring compatibility with bidirectional scanning. As a result, only pixels corresponding to the binary mask that belong to the neuronal pixel were illuminated, substantially reducing the average excitation power. For functional imaging in awake animals, particularly in deep brain regions, motion artifacts cannot be neglected. To introduce tolerance in spatial targeting, the Cellpose-generated masks were dilated by a defined number of pixels. For example, Fig. 6e shows expansion of the ROI mask by 2 pixels along the perimeter, accommodating motion, while preserving substantial power savings. We note that under conditions where the power needs for generating the full-field neuronal activity imaging for generation of masks should be prohibitively high, masks can be also reliably generated by using longer time-integrated low-SNR recordings at sufficiently lower excitation powers, as they only provide initial template spatial segmentation.

Simulation of three-photon fluorescence signal attenuation at high excitation NA

To investigate how excitation NA and tissue scattering affect 3p fluorescence signal generation, we performed Monte-Carlo simulations that modeled ballistic photon propagation followed by nonlinear 3p signal estimation. The simulation assumes a homogeneous scattering medium with an EAL of 300 µm at an excitation wavelength of 1,300 nm. The optical system simulated included a paraxial objective lens with a focal length of 12.5 mm (equivalent to focal length of our 16× Nikon objective) and an immersion medium with refractive index of 1.33. The initial positions of the photons were drawn from a Gaussian distribution at the back-focal plane (BFP), width Gaussian widths chosen to result in a range of excitation NAs (0.2–1.0). Photon trajectories were simulated using a custom Monte-Carlo model that launches photons from the BFP and propagates them through the lens and into the tissue. Each photon is directed such that it would reach the geometric focal plane in the absence of scattering. Within the scattering tissue the photons are assigned a travel distance drawn from an exponential distribution characterized by the EAL and for a given focal depth. Photons that reached the focal plane without being scattered were accumulated as ballistic photons and thus the fraction of ballistic photons was computed for each depth and excitation NA.

To quantify the 3p fluorescence signal, we calculated two scenarios: one that ignores angular-dependent marginal ray attenuation (thus, the loss of photons only depends on the focal depth), and one that incorporates scattering-induced losses of the marginal rays using the Monte-Carlo results. In the former case, the pulse energy at the sample is compensated for the depth-dependent exponential loss, making the 3p signal independent of focal depth, but increasing with the excitation NA as expected. In the latter case, identical pulse energy compensation, as in the former case, was also applied for the given depth and the fraction of ballistic photons from the Monte-Carlo simulation was used to account for the loss of marginal rays, which becomes substantial at high excitation NAs. In both cases, the resulting nonlinear 3p fluorescence signals were normalized to the maximum signal for the case without marginal ray attenuation for comparison.

Data analysis pipeline

The raw tiff data obtained from ScanImage were de-interleaved to extract channels of interest, followed by residual line phase offset

correction due to the bidirectional scan along the fast axis. For long recordings, motion correction was performed using the NoRMCorre library[76]. We used the following parameters for the rigid motion correction (see NoRMCorre documentation): upsampling factor of 50, maximum shift of 15 pixels, initial batch size of 200 and bin width of 200. Other parameters were kept at their defaults. Example of motion correction metrics are shown in Extended Data Fig. 6 for two datasets at imaging depths 730 µm and 1,840 µm (corresponding to the single-plane Ca^{2+} imaging data in Fig. 4), indicating good registration. For 3D volume reconstruction of functional imaging data, mean projections were calculated from single-plane data with an axial separation of 5–20 µm. Spatial components of neuron ROIs were extracted using Cellpose in MATLAB[74]. The individual planes were then combined into single volumes (Figs. 4b and 5b). For structural imaging with GFP mice, 3D volumes were reconstructed from the raw data to keep the fine structures of the neurons, as shown in Fig. 3.

To extract neuron spatial footprints and deconvolved activity traces, we used the CNMF package CaImAn[64] that performs spatiotemporal filtering and constrained non-negative matrix factorization to find components. We applied a rolling mean filter of window size 5 frames on the motion- and phase-corrected datasets before running it through CaImAn pipeline. Key CaImAn settings included (see CaImAn documentation): initialization method 'greedy', search method 'ellipse', deconvolution 'constrained FOOPSI-SPGL1', autoregression 'decay', estimated ROI width 8 pixels, marge threshold 0.9, fudge factor 0.99, frame rate 8.6 and min. SNR 1.5, temporal iterations 5. Other parameters were kept at their defaults. The components were manually classified based on the ROI refinement feature of the CaImAn-based EZcalcium[77] toolbox with parameters mean width (5–10 pixel), roundness (>0.8), oblongness (<2) and sorted by activity (*z*-score). The results are shown in Figs. 4 and 5 and Extended Data Figs. 7 and 8. For all time-projected images shown, the contrast or color saturation was adjusted to aid visualization.

Immunohistochemical validation for the usable average laser power

To determine the safe limit of laser exposure at the sample for in vivo imaging with TrUDI, histological labeling was performed on the brain tissue subjected to the same experimental condition as the GCaMP imaging. All laser exposure experiments were conducted with awake head-fixed wild-type mice at least 2 weeks after implantation of an 8-mm cranial window. For each exposure experiment, a FOV area of ~230 × 230 μm^2 was continuously scanned at 8.6 Hz for ~10 min at a depth of 2 mm below the brain surface with an average laser power 150, 200 or 250 mW at the objective. For the negative control condition, the cranial window was installed but no laser exposure occurred.

At 16 h following imaging, mice were deeply anesthetized with isoflurane and transcardially perfused with cold phosphate-buffered saline (PBS) followed by 4% paraformaldehyde (PFA). Brains were extracted, post-fixed in 4% PFA for 24 h and then transferred to 30% sucrose in PBS at 4 °C. Coronal sections (50-µm thickness) were collected from regions within and surrounding the imaging FOV using a cryostat.

For astrocyte activation labeling, free-floating sections were permeabilized in 0.3% Triton X-100 in PBS (PBS-T), followed by blocking in a solution containing 5% normal goat serum (NGS) and 1% bovine serum albumin (BSA). Sections were incubated with primary mouse anti-GFAP antibody (1:1,000, Proteintech, 60190-1-Ig) overnight at 4 °C under gentle rocking. After washing in PBS, sections were incubated with Alexa Fluor 594-conjugated goat anti-mouse secondary antibody for 2 h at room temperature under gentle rocking. Sections were washed again in PBS, with Hoechst 33342 (Invitrogen, H3570) included during the final wash, before mounting and cover-slipping using antifade mounting medium.

To assess free radical associated DNA damage and neuronal integrity within the same sections, γH2AX immunohistochemistry was performed in combination with NeuroTrace labeling. Free-floating sections were permeabilized in 0.3% PBS-T and blocked in 5% NGS and 1% BSA. Sections were incubated with primary mouse anti-γH2AX antibody (Thermo Fisher Scientific, MA1-2022; 1:200 dilution) for 36 h at 4 °C under gentle rocking. Following primary antibody incubation, sections were washed in PBS and then incubated with Alexa Fluor 488-conjugated goat anti-mouse secondary antibody for 2 h at room temperature under gentle rocking. After secondary antibody incubation, sections were thoroughly washed in PBS to remove blocking solution and unbound antibody. Sections were then stained with NeuroTrace Red fluorescent Nissl stain (1:100 dilution, Thermo Fisher Scientific, N21482) for 1 h at room temperature. Sections were washed in PBS, with Hoechst 33342 (Invitrogen, H3570) included during the final wash, before mounting and cover-slipping.

Stained sections were imaged using a laser scanning confocal microscope (Zeiss LSM 880, software Zen 2.3) with 10× and 20× objectives for overview imaging and a 63× objective for high-resolution imaging. Image stacks were acquired with a *z*-step size of 2 µm. Quantification of the images were performed as follows using Fiji/ImageJ (1.54p/2.16.0) software.

For GFAP analysis, maximum intensity projections were generated from *z*-stacks. Mean fluorescence intensity was quantified within a 1 × 1 mm^2 ROI centered within the imaging FOV, positioned directly above the CC while avoiding the brain surface. For each condition, measurements from three biological replicates were compared to the 95% confidence interval of the negative control condition (Extended Data Fig. 9d). Statistical significance between each experimental condition and the negative control condition was assessed using a one-sided Welch's *t*-test to test for increases in mean fluorescence response and the *P* values are reported in the corresponding Results section.

For NeuroTrace analysis, neuronal cell density was quantified within a 1 × 1 mm^2 ROI defined as above. Images from the NeuroTrace channel were thresholded to generate a binary mask in Fiji, and particles were identified using the 'Analyze Particles' function with an area range of 70–700 μm^2. The number of positively labeled cells within the ROI was counted, and cell densities were compared across conditions relative to the negative control (Extended Data Fig. 9e). A similar one-sided Welch's *t*-test for statistical significance was assessed to test for decreases in mean cell densities compared to negative control condition and the *P* values are reported in the corresponding Results section.

For γH2AX analysis, a region within the 250-mW condition exhibiting reduced NeuroTrace-labeled cell density was first identified. Corresponding regions were selected within each imaging condition for comparison. γH2AX fluorescence was quantified in two ways: (1) mean fluorescence intensity within a 100 × 100 μm^2 ROI imaged at 63× magnification (Extended Data Fig. 9f), and (2) nuclear fluorescence intensity measured within ROIs defined by segmentation of nuclei using the Hoechst channel. For the latter, nuclei were segmented to generate intranuclear ROIs, and the mean Alexa Fluor 488 (γH2AX) fluorescence intensity was measured within each nucleus (Extended Data Fig. 9g). In both cases, similar test for statistical significance were performed as in the case of GFAP analysis, to test for increases in mean fluorescence response and the *P* values are reported in the corresponding Results section.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The raw data are available from the corresponding author upon reasonable request as the data size is too large for sharing with public repositories. Source data are provided with this paper.

Code availability

Neuronal segmentations based on anatomical or spatiotemporal feature, and motion-corrections were implemented in MATLAB (R2024b) using open-source packages and are publicly available. All custom codes used for data analysis in this work are provided as a compressed archive and are available with the paper as Supplementary Code 1 and can also be accessed from GitHub at https://github.com/vazirilab/TrUDI-ultradeep-3p, under an open-source license permitting not-for-profit research use (a License.md file is available with the code repository).

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Author contributions

K.S. contributed to the conceptualization of the project, built and characterized the imaging system, implemented the data-acquisition system, programmed experimental control and analysis software, performed experiments, analyzed data and wrote the paper. K.B. performed immunohistochemical experiments and analysis. F.L. and M.A.F. performed cranial window surgeries and F.L. contributed to performing experiments. A.B. and J.L. provided additional transgenic animals with cranial windows. A.V. conceived and led the project, conceptualized, designed and guided the implementation of the imaging system and data analysis and wrote the paper.

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Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41592-026-03251-2.

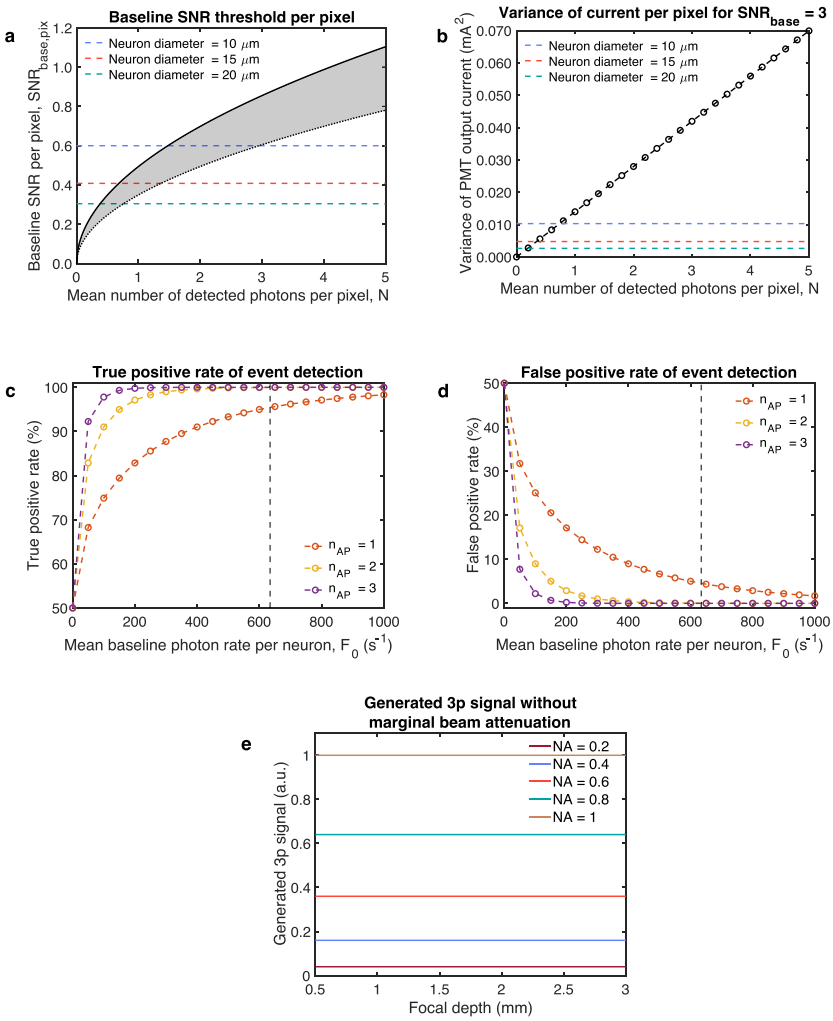
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Peer review information *Nature Methods* thanks Chris Schaffer, who co-reviewed with Aidan Hughes; and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Nina Vogt, in collaboration with the *Nature Methods* team.

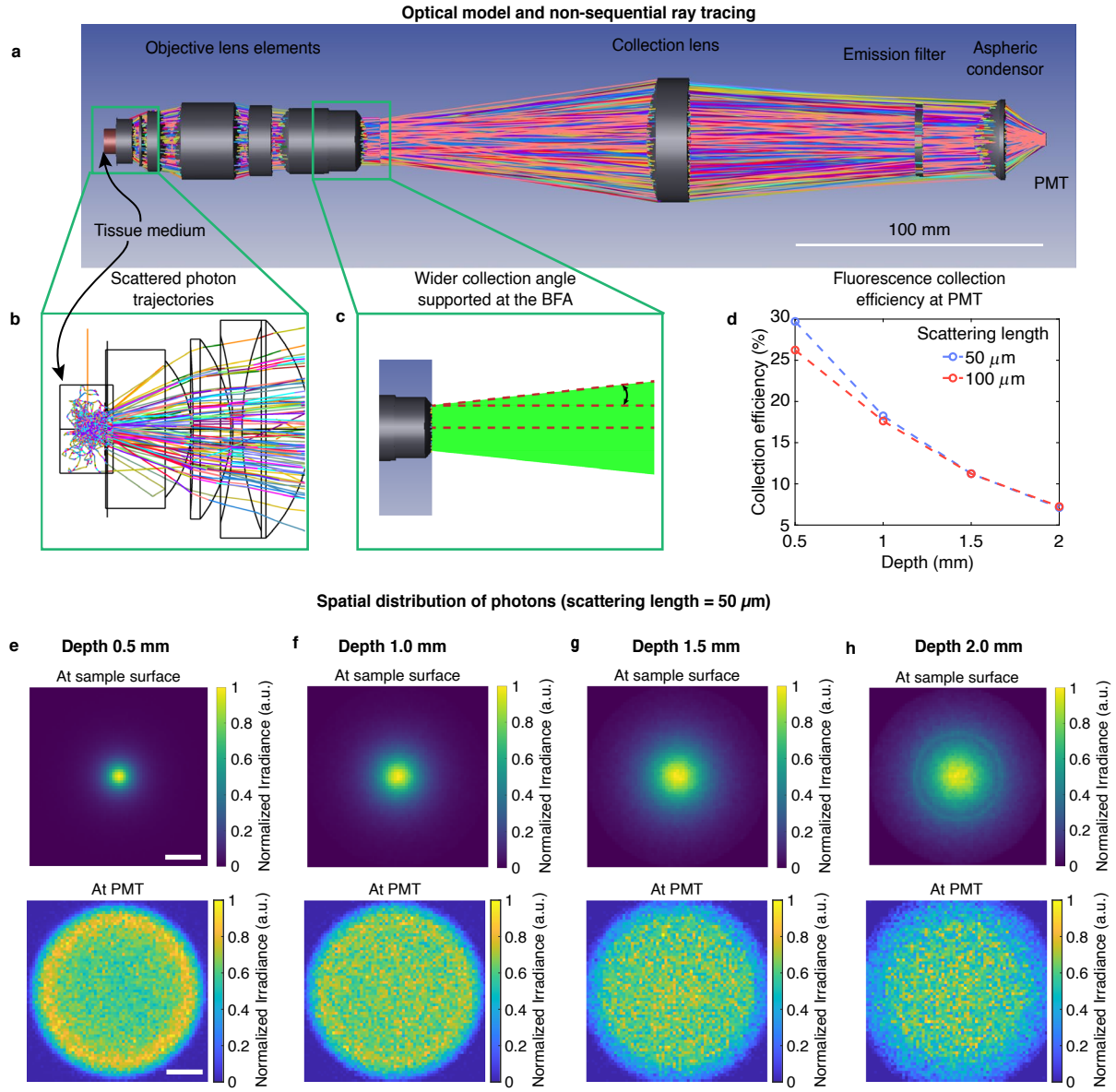
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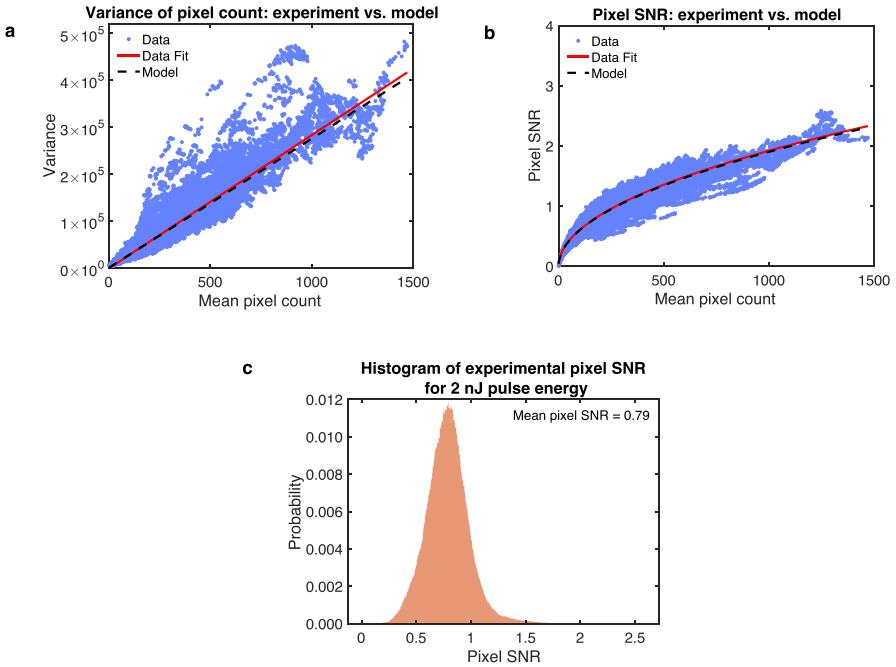
Extended Data Fig. 1 | Baseline SNR thresholds, neuron activity detection metrics, and NA-dependent fluorescence signals generated in 3p microscopy corresponding to the optimal parameter choices. (**a-b**) Baseline SNR thresholds and variance of the PMT output current per pixel: (**a**) Baseline SNR per pixel ($SNR_{base,pix}$) as functions of the mean number of detected photons per pixel at the PMT. The plot for $SNR_{base,pix}$ is shown for both conditions where background is negligible (that is, SBR » 1, upper solid line of the shaded region) as well as at SBR-limited depth (that is, SBR = 1, lower dotted line of the shaded region). The horizontal dashed lines represent minimum threshold for baseline SNR per pixel to achieve baseline SNR of 3 per neuron for three different neuron sizes with diameters of 10, 15, and 20 µm. (**b**) Variance of PMT output current as functions of the mean number of detected photons per pixel at the PMT. The horizontal dashed lines represent the threshold to achieve baseline SNR of 3 per neuron for three different neuron sizes with diameters of 10, 15, and 20 µm. (**c**,**d**) Detection accuracy indicators of an underlying neuronal activity event at the SBR-limited depth (SBR = 1): (**c**) True positive rate (TPR) and (**d**) false positive rate (FPR) are plotted as a function of mean baseline photon rate per neuron (per second). In **c**,**d**, the vertical dashed lines represent the mean baseline photon rate per neuron sampled with 25 pixels at a rate of 8.6 Hz in order to achieve a minimum baseline SNR threshold of 3 per neuron. (**e**) Generated 3p fluorescence signal as a function of focal depth for different excitation NAs under the assumption that scattering-related photon loss depends solely on depth, and attenuation of marginal rays is ignored. Note: Pulse energy at the sample is compensated for the depth-dependent exponential loss, making the 3p signal independent of focal depth but increasing with the excitation NA.

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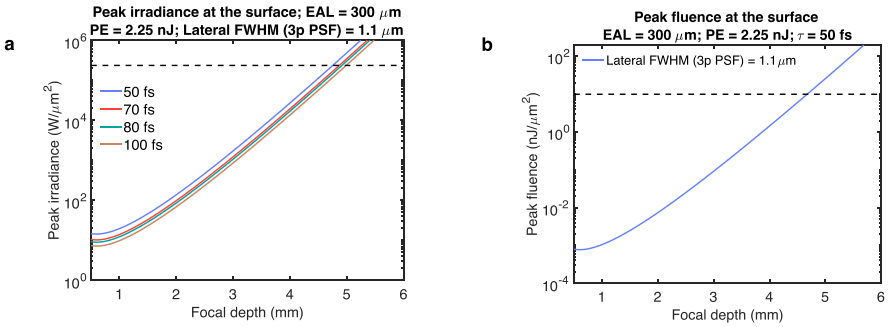
Extended Data Fig. 2 | Optical model and collection efficiency of fluorescence detection path in TrUDI. (**a**) Zemax model of the full detection path in TrUDI consisting of a reconstructed approximate model for the objective lens of Nikon, CFI75 LWD 16X W, with nominal collection NA of 0.8 and working distance 3 mm. Key components in the collection system consists of an achromatic doublet (Edmund Optics, NT49-284-INK, $f = 100$ mm, 50 mm dia.) and aspheric condenser lenses (Newport, KPA034-C, 28.5 mm EFL, 32.5 mm dia.). The detector of active area 5 mm × 5 mm represents the Hamamatsu PMT (H10770-PA40). A few representative non-sequential rays reaching to the PMT are shown (color represents each segment of rays after reflection or refractions). Scale bar 100 mm. (**b**) A zoom-in into the tissue volume region showing a subset of ray paths in non-sequential ray tracing mode followed from Monte-Carlo light scattering from a source at the tissue depth of 2 mm at the peak of the fluorescence emission wavelength of 515 nm (Supplementary Note 5). (**c**) A schematic view of effective collection angle of the optical system which allow up to ~8-degree wider collection angle in this design compared to collimated configuration at the back-focal plane. (**d**) Collection efficiency (the ratio of power received that the PMT detector to power at the source located at the center of the field of view at a given depth) as a function of depth for scattering lengths of 50 μm and 100 μm, consistent with the typically reported range for brain tissues in literature[3–6]. (**e**-**h**) Distribution of photons reaching the sample surface (top) and the PMT (bottom) for a fluorescence source placed at depths of 0.5, 1, 1.5 and 2.0 mm, respectively and scattering length of 50 μm. Scale bars, 2 mm (top), 1 mm (bottom).

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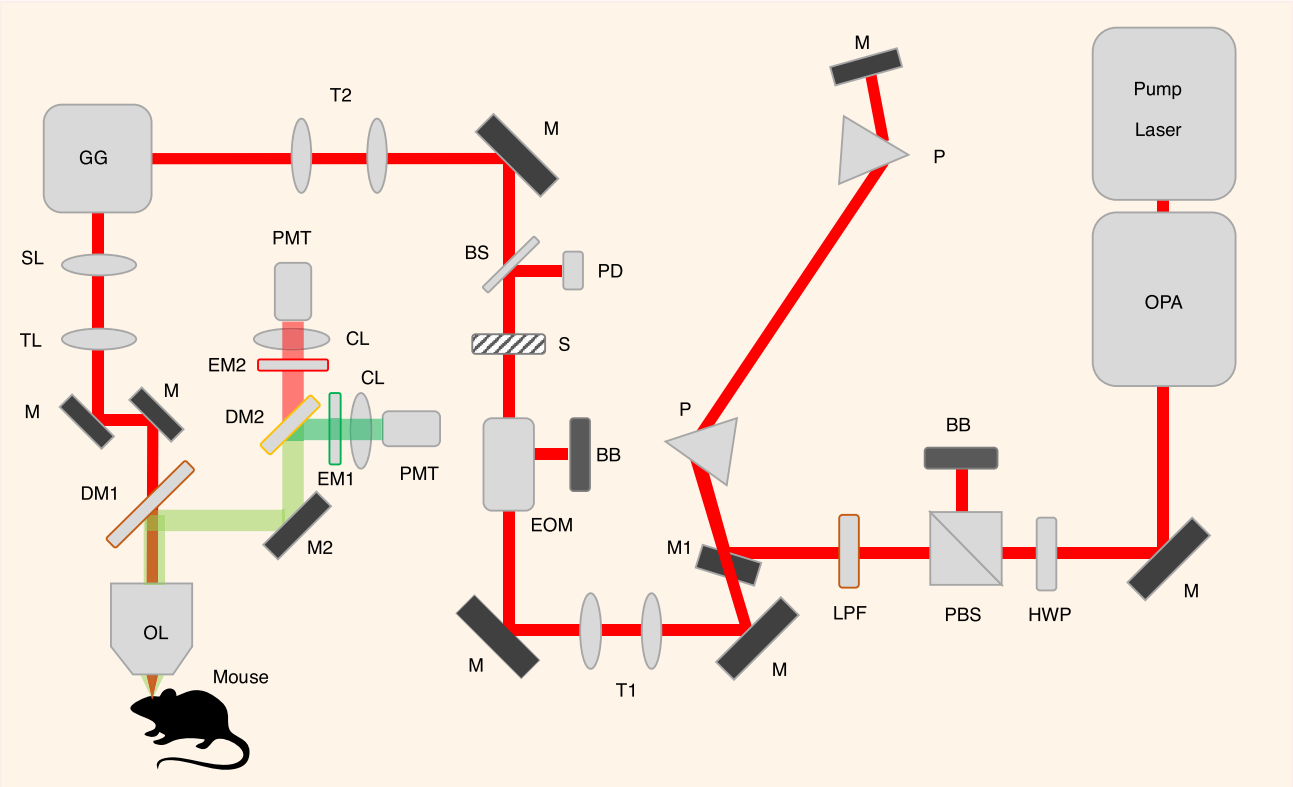
Extended Data Fig. 3 | Experimental evaluation of system noise and SNR and comparison with theoretical SNR model in three-photon microscopy. (**a**) Variance of pixel count as a function of mean pixel count from raw image data. The slope of the linear fit (red solid line) is ~280, and the fit agrees well with the theoretical model for variance (green dashed line) (Supplementary Note 3). (**b**) Pixel SNR as a function of mean pixel count from raw image data. A power-law, $y = Ax^B$, fits the data well, with coefficients $A = 0.06$ and $B = 0.5$, in good agreement with the theoretical SNR model (green dashed line) (Supplementary Note 3). (**c**) Histogram of experimental pixel SNR over the FOV across multiple 100-frame windows. The experimental data were collected under the typical conditions used in our imaging system, that is, FOV of 230 × 230 μm^2; pixel size of 1.8 × 1.8 μm^2 at an 8.6 Hz frame rate. We used ~2 nJ pulse energy at the focus using the single-pulse-per-pixel data acquisition scheme. To estimate pixel statistics, we used a 100-frame window. The theoretical model and the parameters are given in Supplementary Note 3.

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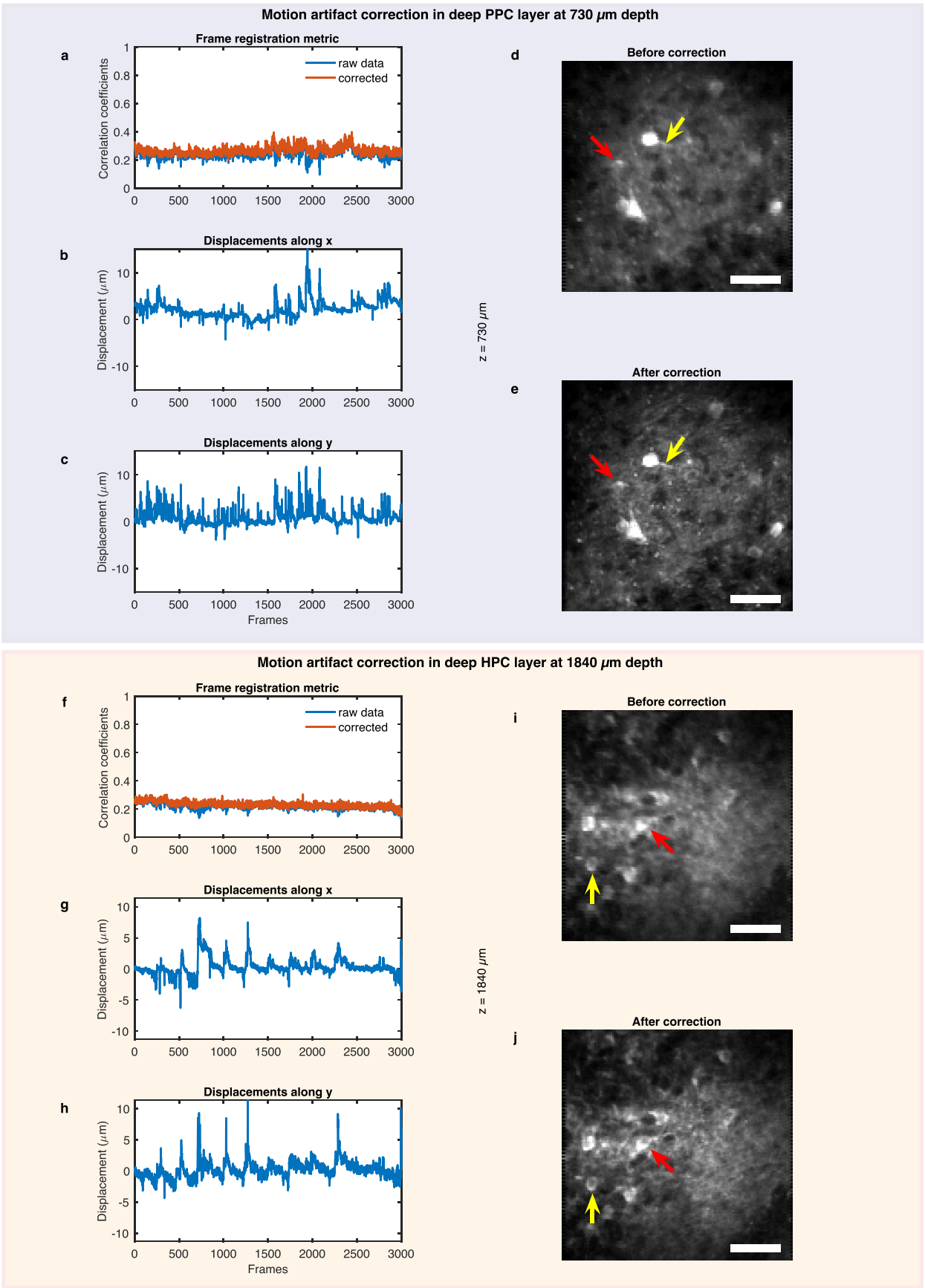
Extended Data Fig. 4 | Laser-induced nonlinear damage thresholds at the tissue surface for hypothetical extended depth reach in three-photon microscopy. a) The peak irradiance at the surface as a function of focal depth extended up to a hypothetical depth of 6 mm for a pulse energy (PE) of 2.25 nJ, lateral 3p PSF size of 1.1 µm (FWHM), and an EAL of 300 µm at 1300 nm. Each colored line corresponds to a different pulse duration. The horizontal dashed line corresponds to the peak irradiance limit of ~2.35×10^5 W/µm² for the safe exposure[13]. (**b**) The peak fluence at the surface as a function of focal depth extended up to a hypothetical depth of 6 mm for a lateral 3p PSF size of 1.1 µm (FWHM) with PE of 2.25 nJ, pulse duration of 50 fs (FWHM), and an EAL of 300 µm at 1300 nm. The horizontal dashed line corresponds to the peak fluence limit of 10 nJ/µm² for the safe exposure[13].

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Extended Data Fig. 5 | Schematic of the optical setup for Three-photon Ultra-Deep Imaging (TrUDI). TrUDI microscope beam path schematic with a galvo-galvo scanner, a power control module and a dispersion compensation module. Pump Laser (Light-Conversion, Carbide CB3 80 W); OPA (Light-Conversion, Orpheus-F); M, mirror (Flat silver mirror Thorlabs PF10-03-P01 or similar); HWP, half-waveplate (Thorlabs AHWP10M-1600); PBS, polarizing beamsplitter (Thorlabs PBS124); BB, beam-block; LPF, 900 nm long-pass filter (Thorlabs FELH0900); M1, D-shaped mirror for beam pick-up (Thorlabs PFD10-03-P01); P, prism-pair for dispersion compensation (Newport 10SF10); T1–T2, telescope to resize laser beam; EOM, electro-optical modulator for fast power adjustment (Conoptics Model 160-40); S, laser shutter (Thorlabs SH05/M); BS, beam sampler (Thorlabs BSF05-C); PD, photodiode (Thorlabs PDA20CS2); GG, galvo-galvo scanner, 5 mm (Novanta, 6215H XY scanner); SL, scan lens f = 50 mm (Thorlabs SL50-3P); TL, tube lens, f = 200 mm (Thorlabs TL200-3P); Sutter 3p-MOM detection optics (DM1, Semrock FF735-Di02-25×36; M2, elliptical curved mirror; DM2, Chroma 565DCXR; EM1, Chroma ET525/70 m; EM2, Chroma ET605/70 m; CL, condenser collection lenses; PMT, Hamamatsu H10770PA-40); OL, objective lens (Nikon 16×, 0.8 NA, 3 mm WD, CFI75 LWD 16X W).

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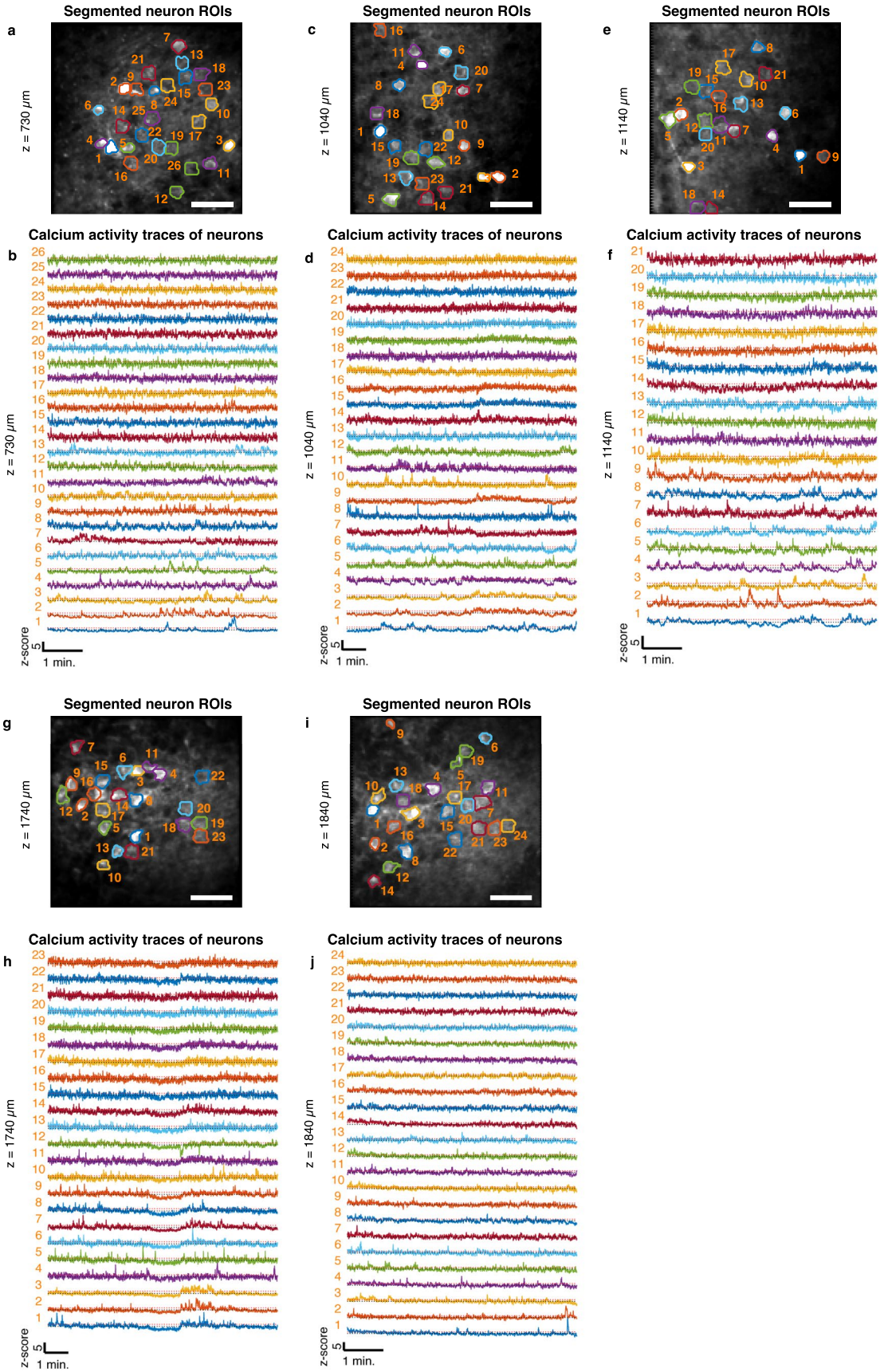


Extended Data Fig. 6 | See next page for caption.

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Extended Data Fig. 6 | Estimation of motion artifacts and their correction in deep 3p imaging in the mouse brain on the posterior parietal cortex (PPC) and hippocampus (HPC). (**a-e**) Motion artifact correction in deep PPC layer at 730 µm depth. Correction metrics: (**a**) Correlation coefficient as frame registration metric for raw and motion-corrected data. (**b-c**) Detected displacements along x and y in a single plane caused by motion, respectively. (**d-e**) Mean projection of the single-plane recording before and after motion correction, respectively. The red and the yellow arrows highlights reduced motion blur at two locations. Scale bar, 50 µm. (**f-j**) Motion artifact correction in a deep HPC layer at 1840 µm depth. Correction metrics: (**f**) Correlation coefficient as frame registration metric for raw and motion-corrected data. (**g-h**) Detected displacements along x and y in a single plane caused by motion, respectively. (**i-j**) Mean projection of the single-plane recording before and after motion correction, respectively. The red and the yellow arrows highlights reduced motion blur at two locations. Scale bar, 50 µm.

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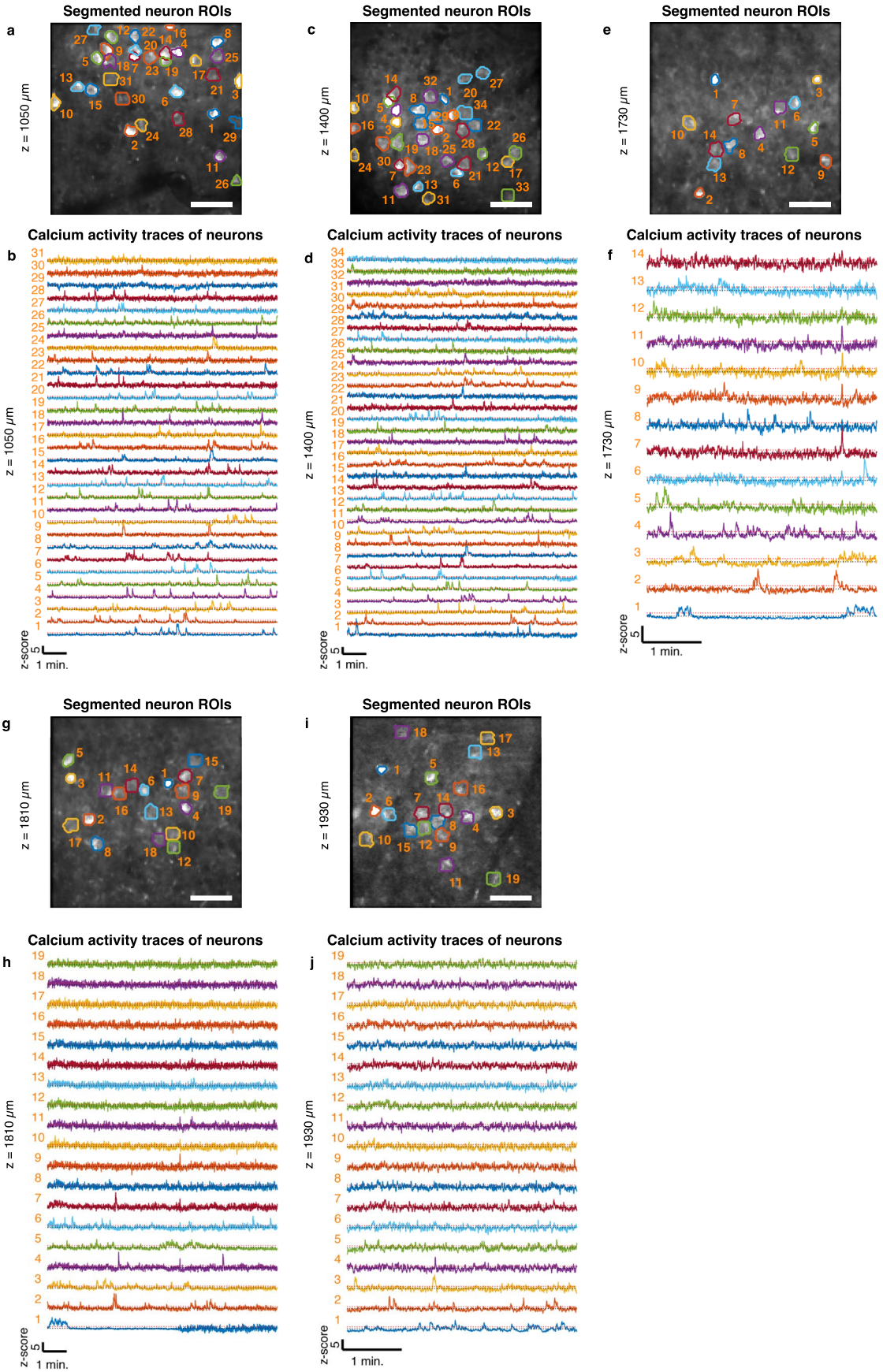


Extended Data Fig. 7 | See next page for caption.

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Extended Data Fig. 7 | All segmented neurons and corresponding CaImAn-extracted calcium (Ca^{2+}) activity traces across depths in posterior parietal cortex (PPC) and hippocampus (HPC). (**a, c, e, g, i**) Mean intensity projection of neuroactivity in individual planes covering a depth range 730–1840 µm. Segmented neurons are labeled in sequence based on their activity (z-score). (**b, d, f, h, j**) Ca^{2+} activity traces of the neurons in (**a, c, e, g, i**), respectively, over ~6–10 min. using CaImAn[14] with z-scores up to ~10 for the strongest responding neurons. The red dotted line denotes z-score = 1 and the black dotted line denotes the baseline. The average power used for each of the five depths was 2.5, 6.5, 12, 58, and 66 mW, respectively. Imaging area, 230 × 230 μm^2; pixel size, 1.8 × 1.8 μm^2. Scale bar, 50 µm.

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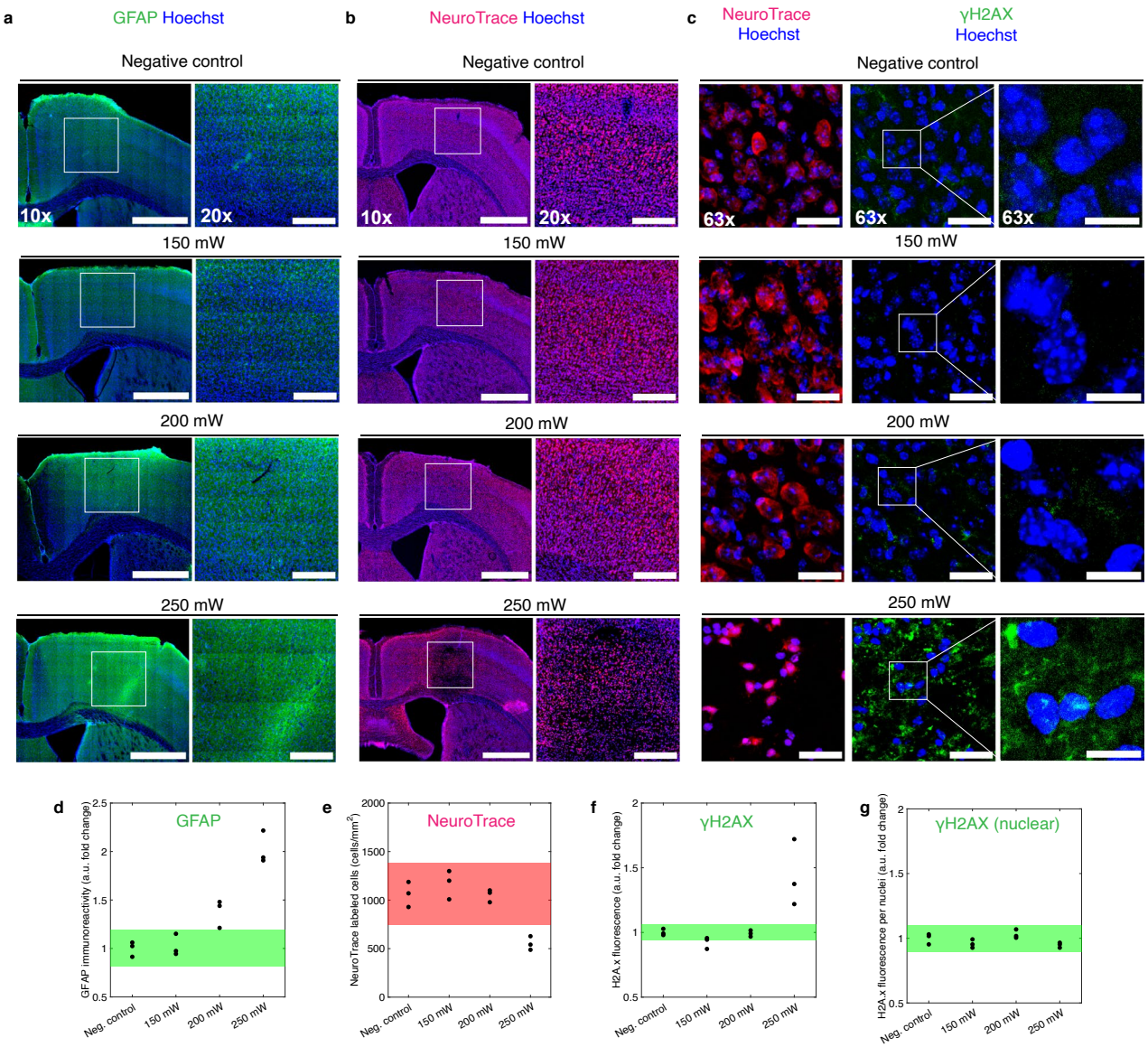


Extended Data Fig. 8 | See next page for caption.

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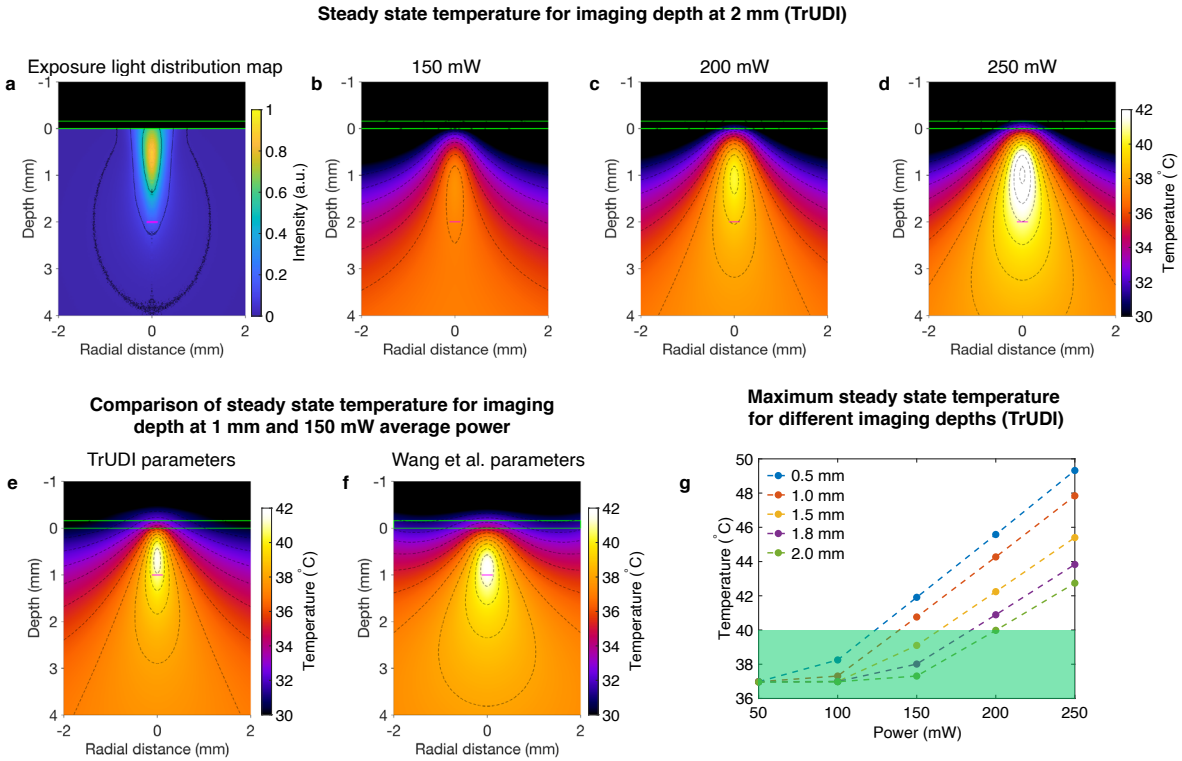
Extended Data Fig. 8 | All segmented neurons and corresponding CaImAn-extracted calcium (Ca^{2+}) activity traces across depths within motor cortical areas and the striatum. (**a, c, e, g, i**) Mean intensity projection of neuroactivity in individual planes covering a depth range 1050–1930 µm. Segmented neurons are labeled in sequence based on their activity (z-score). (**b, d, f, h, j**) Ca^{2+} activity traces of the neurons in (**a, c, e, g, i**), respectively, over ~4–10 min using CaImAn[14] with z-scores up to ~12 for the strongest responding neurons. The red dotted line denotes z-score = 1 and the black dotted line denotes the baseline. The average power used for each of the five depths was 10, 30, 120, 120, and 120 mW, respectively. Imaging area, 230 × 230 μm^2; pixel size, 1.8 × 1.8 μm^2. Scale bar, 50 µm.

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Extended Data Fig. 9 | Experimental validation of linear and nonlinear safe exposure limit in deep imaging using TrUDI. Histological markers of astrocytic activation, neuronal integrity, and DNA damage following laser exposure at 2 mm depth. (**a**) Representative overlayed images of coronal brain sections with GFAP (astrocyte activation, green) and Hoechst (nuclei, blue) staining acquired at 10× (left) and 20× (right) magnification across conditions: negative control (top row), experimental condition with 150 mW, 200 mW, and positive control with 250 mW (bottom row). Boxes in the 10× images indicate ~1×1 mm^2 regions of interest (ROIs) shown at higher magnification in the adjacent 20× images used for quantification of relative fluorescence intensity. Scale bars, 1 mm (10×) and 300 µm (20×). (**b**) Representative overlayed images of coronal brain sections with NeuroTrace (neuronal somata, red) and Hoechst (nuclei, blue) staining across the same conditions and layout as in (a). Boxes in the 10× images indicate ~1×1 mm^2 ROIs shown at higher magnification in the adjacent 20× images used for quantification of cell density. Scale bars, 1 mm (10×) and 300 µm (20×). (**c**) Overlayed images from coronal brain sections at high-magnification (63×) showing NeuroTrace (red) and Hoechst (blue) on the left, γH2AX (green) and Hoechst (blue) at the center, and corresponding zoomed-in view on the right, staining across the same conditions (top to bottom: negative control, experimental condition at 150 mW, 200 mW, 250 mW). Scale bars, 30 µm for the left and the center, and 10 µm for the zoomed view on the right. (**d**) Quantification of the GFAP immunoreactivity: Mean fluorescence intensity over a ~1×1 mm^2 area normalized to the group mean of the same for negative control group. N = 3 for all exposure conditions; Shaded area denotes the 95% confidence interval of the control group mean. (**e**) Quantification of the NeuroTrace-labeled cells: Number of cells counted over a ~1×1 mm^2 area. N = 3 for all exposure conditions; Shaded area denotes the 95% confidence interval of the control group mean. (**f-g**) Quantification of the γH2AX fluorescence: (f) Mean fluorescence intensity over a ~100 × 100 $µm^2$ area normalized to the group mean of the same for negative control group. (g) Mean fluorescence intensity within nuclear ROIs area marked by Hoechst normalized to the group mean of the same for negative control group. N = 3 for all exposure conditions; Shaded areas denote the 95% confidence interval of the control group mean. For all laser exposure experiments: Imaging depth 2 mm, scan area, 230 × 230 $µm^2$; frame rate 8.6 Hz, duration ~10 min., single pulse per pixel. Negative control received no laser exposure.

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Extended Data Fig. 10 | Simulation of temperature distribution within brain tissue for different laser power exposures at 1300 nm. (**a**-**d**) Distribution of light intensity (**a**), and the steady state temperature maps, (**b**-**d**) for exposure with 1300 nm light focused the imaging depth of 2 mm for different average powers (150, 200, and 250 mW) out of the objective in TrUDI. The contour lines in (**b**-**d**) represent the isotherms with 1 °C separation with highest contour-line temperature of 37 °C in (**b**), 40 °C in (**c**) and 43 °C in (**d**). (**e**) Temperature maps for the imaging depth of 1 mm at an average power of 150 mW out of the objective but otherwise based on imaging parameters and conditions as used for 2 mm imaging in this work. (**f**) Temperature maps for the imaging depth of 1 mm at an average power of 150 mW out of the objective based on imaging parameters and conditions used in Wang et al.[10], demonstrating similar degree of increase in steady state temperature. The contour lines in (**e**) and (**f**) represent the isotherms with 1 °C separation with the highest contour-line temperature of 41 °C consistent with both, temperature modeling results and experimental observation of the onset of immunohistochemical response in Wang et al.[10] under these conditions. (**g**) Plot of maximum induced steady state temperature (°C) in the tissue volume as a function of average power out of the objective for different imaging depths. The maximum temperature is calculated as the highest average temperature within a volume 0.01 mm^3 (Supplementary Note 6). The green band represent the temperature < 40 °C. For all simulation results for TrUDI, (a-e) and (g) the following key simulation parameters were used: Scan field of view (FOV), 230 × 230 μm^2 (magenta line); Effective excitation, NA ~ 0.3; Brain core temperature, 37 °C; Water temperature above the cranial window (green line), 21 °C equal to the laboratory room temperature (See Supplementary Note 6 for more details). For simulation result based on Wang et al., (**f**), the key differentiating parameters were: Wavelength, 1320 nm; NA, ~ 0.75; Water temperature, 25 °C (See Supplementary Note 6 for details).

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☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Data was collected using ScanImage 2022 by MBF Bioscience. Histology experimental data was collected with Zeiss Zen 2.3.
Data analysis	Neuronal segmentations based on anatomical or spatio-temporal features, as well as motion corrections, are implemented in MATLAB (R2024b) using the open-source packages Cellpose, CaImAn, and NoRMCorre, respectively, as described in the Methods section. Any custom data analysis code described in the Methods section for pre- or post-processing is available as supplementary file with the manuscript. Histology data has been analyzed in Fiji/imageJ (1.54p/2.16.0).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All source data supporting the findings of the study are provided with the paper. The raw data are available from the corresponding author upon reasonable request.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	*Use the terms sex (biological attribute) and gender (shaped by social and cultural circumstances) carefully in order to avoid confusing both terms. Indicate if findings apply to only one sex or gender; describe whether sex and gender were considered in study design; whether sex and/or gender was determined based on self-reporting or assigned and methods used. Provide in the source data disaggregated sex and gender data, where this information has been collected, and if consent has been obtained for sharing of individual-level data; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex- and gender-based analyses where performed, justify reasons for lack of sex- and gender-based analysis.*
Reporting on race, ethnicity, or other socially relevant groupings	*Please specify the socially constructed or socially relevant categorization variable(s) used in your manuscript and explain why they were used. Please note that such variables should not be used as proxies for other socially constructed/relevant variables (for example, race or ethnicity should not be used as a proxy for socioeconomic status). Provide clear definitions of the relevant terms used, how they were provided (by the participants/respondents, the researchers, or third parties), and the method(s) used to classify people into the different categories (e.g. self-report, census or administrative data, social media data, etc.) Please provide details about how you controlled for confounding variables in your analyses.*
Population characteristics	*Describe the covariate-relevant population characteristics of the human research participants (e.g. age, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."*
Recruitment	*Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.*
Ethics oversight	*Identify the organization(s) that approved the study protocol.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes, in our case the number of mice and recordings, were chosen to ensure animal-to-animal and recording-to-recording variability was reflected in the captured data. We imaged a minimum of N = 3 mice with 5-10 different recordings for each in the study. This minimum criterion is consistent with other reports in the literature. Additionally, the focus of this paper is to introduce a principled method and its technological advancement rather than provide biological conclusions, thus larger sample sizes were deemed unnecessary.
Data exclusions	No data were excluded from analysis.
Replication	Experiments and analysis were replicated across the mice and recordings used in these study.
Randomization	Each animal subject was considered it's own group, therefore randomization of animals into testing groups was not applicable to this study.
Blinding	This study did not feature direct testing of hypotheses during experiments, therefore researcher bias was not expected to affect data collection and blinding was deemed unnecessary.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Mouse anti-GFAP antibody (1:1000). Vendor: Proteintech. Species: Mouse. Cat. No.: 60190-1-Ig. Clone No.: 4B2E10. Lot No.: 1004401

Alexa Fluor 594–conjugated goat anti-mouse

Phospho-Histone H2A.X (Ser139) monoclonal antibody (1:200). Vendor: Thermo Fisher Scientific. Species: Mouse. Cat. No.: MA1-2022. Clone No.: 3F2. Lot No.: 3303669

Alexa Fluor 488–conjugated goat anti-mouse secondary

Validation

Both antibodys were validated by the manufacturer

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals

Experiments used mice of 4-18 weeks of age of the following strains:Thy1-EGFP (The Jackson Laboratory, Strain #: 007788), Camk2a-tTA (The Jackson Laboratory, Strain #: 007004) × tetO-GCaMP6s (The Jackson Laboratory, Strain #: 024742), Camk2a-tTA (The Jackson Laboratory, Strain #: 016198) × tetO-GCaMP8s (The Jackson Laboratory, Strain #: 037717), and VGAT-Cre (The Jackson Laboratory, Strain #: 016198) × GCaMP8m (The Jackson Laboratory, Strain #: 037718), and C57BL/6J (The Jackson Laboratory, Strain #: 000664). . Mouse were housed in pairs on a reverse dark/light cycle (8am-8pm dark, 8pm-8am light) at normal room temperature and humidity.

Wild animals

This study did not involve wild animals.

Reporting on sex

Experiments used male and female mice.

Field-collected samples

This study did not involve field-collected samples.

Ethics oversight

All surgical and experimental procedures were approved by the Institutional Animal Care and Use Committee of The Rockefeller University.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	NA
Novel plant genotypes	NA
Authentication	NA